ELITE IGCSE ACADEMY

Private Teacher Edition

Elite IGCSE Classified Unit 1 Expertise Answers (Q20+)

127 Questions ▪ Question page then worked-solution page

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2026 EDITION

Unit 1 / 4WM1

Private answer book - Q20+ expertise.

Each question is followed by its worked solution on the next page.

Contents

Topic 1

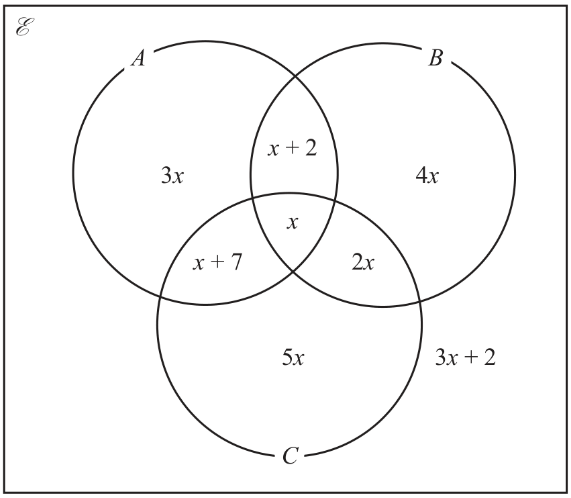
Set Notation & Venn Diagrams

CLASSIFIED PROBLEM Paper: P2H | Session: Nov 2024 | Q#: 21 | Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Source: Nov 2024 P2H

21 The Venn diagram shows information about the numbers of items in set A, set B and set C, where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

$n(A' \cap C) =$ ....

(Total for Question 21 is 4 marks)

WORKED SOLUTION

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Nov 2024 P2H | Question 21

Set Notation & Venn Diagrams

Topic check. Set notation and Venn diagrams. The tag is correct.

Method

The complement of $A \cup B$ contains the regions outside both A and B. These are $5x$ and $3x + 2$.
So $5x + (3x + 2) = 26$, hence $8x + 2 = 26$ and $x = 3$.
Now $A' \cap C$ means inside C, but not inside A. These are the regions $5x$ and $2x$.
So $n(A' \cap C) = 5x + 2x = 7x = 7(3) = 21$.

FINAL ANSWER

21

Topic 2

Rounding, Estimation & Bounds

CLASSIFIED PROBLEM Paper: P1HR | Session: Jan 2022 | Q#: 22 | Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Source: Jan 2022 P1HR

22 The diagram shows triangle ABC

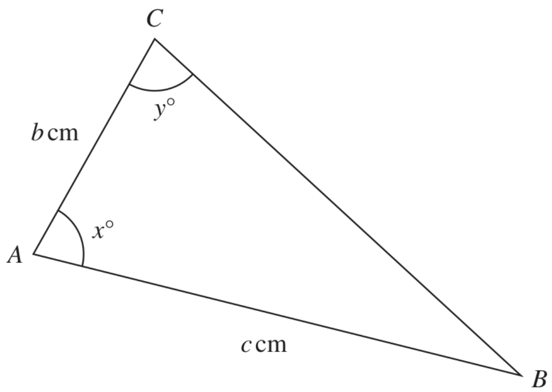


Diagram **NOT** accurately drawn

$c = 11.5$ correct to one decimal place
$x = 80$ correct to the nearest whole number
$y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of b
Show your working clearly.
Give your answer correct to 3 significant figures.

....

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 22
Rounding, Estimation & Bounds

Topic check. Rounding, Estimation & Bounds is correct. The question also uses the sine rule, but the main demand is choosing the correct bounds.

Method

For an upper bound for b, use the upper bound of c:
$c = 11.55$
Use the lower bounds of x and y, because this makes angle B as large as possible and makes $\sin y$ as small as possible.
$x = 79.5°, \quad y = 74.5°$
So
$\angle B = 180° - 79.5° - 74.5° = 26°$
Using the sine rule,

$$\frac{b}{\sin 26°} = \frac{11.55}{\sin 74.5°}$$

$$b = \frac{11.55 \sin 26°}{\sin 74.5°} = 5.254\ldots$$

FINAL ANSWER
5.25 cm to 3 significant figures.

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2023 | Q#: 25 | Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Source: Jun 2023 P2H

25 A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in g/cm^3 correct to 2 decimal places.
Show your working clearly.

.... g/cm^3

(Total for Question 25 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 25
Rounding, Estimation & Bounds

Topic check. Rounding, Estimation & Bounds is correct. The question is about choosing bounds before using the density formula.

Method

For the upper bound of density, use the upper bound of the mass and the lower bound of the volume.
The radius is 2.8 cm correct to 1 decimal place, so the lower bound is
$r = 2.75$
$M = 260$ correct to 2 significant figures, so the upper bound is
$M = 265$
The mass is given as $M\pi$ grams, where $M = 265$ for the upper bound.
The volume of the sphere is

$$\frac{4}{3}\pi r^3$$

So

$$\text{density} = \frac{M\pi}{\frac{4}{3}\pi r^3}$$

The π cancels:

$$\text{density} = \frac{3M}{4r^3}$$

Upper bound:
$\dfrac{3(265)}{4(2.75)^3} = 9.5567...$

FINAL ANSWER
9.56 g/cm^3 to 2 decimal places.

CLASSIFIED PROBLEM

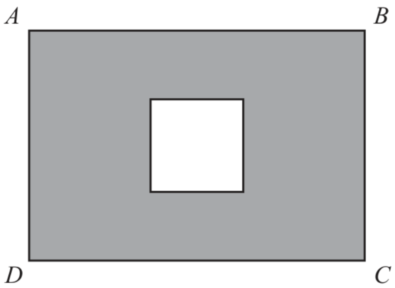
Paper: P2H | Session: Jun 2025 | Q#: 22 | Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Source: Jun 2025 P2H

22 The diagram shows a square inside rectangle *ABCD*

Diagram **NOT** accurately drawn



The total area of the region shown shaded in the diagram is $X\,\text{cm}^2$

$AB = 11.5$ cm correct to the nearest 0.5 cm
$BC = 9.2$ cm correct to 2 significant figures
side of square = 4.1 cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X =$ ....

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 22

Rounding, Estimation & Bounds

Topic check. Rounding, Estimation & Bounds is correct.

Method

The shaded area is

$X =$ area of rectangle $-$ area of square

Bounds:

$11.25 \leq AB < 11.75$

$9.15 \leq BC < 9.25$

$4.05 \leq$ side of square < 4.15

Lower bound for X:

$11.25(9.15) - 4.15^2 = 85.715$

Upper bound for X:

$11.75(9.25) - 4.05^2 = 92.285$

So $85.715 \leq X < 92.285$. Both bounds round to 90 to the nearest 10.

FINAL ANSWER

$X = 90$ cm^2 to a suitable degree of accuracy.

CLASSIFIED PROBLEM

Paper: P1H **Session: May 2021** **Q#: 20** **Marks: 3**

Topic: **Rounding, Estimation & Bounds**

Source: May 2021 P1H

20 $P = \dfrac{t - w}{y}$

t = 9.7 correct to 1 decimal place

w = 5.9 correct to 1 decimal place

y = 3 correct to 1 significant figure

Calculate the upper bound for the value of P.
Show your working clearly.

....

(Total for Question 20 is 3 marks)

WORKED SOLUTION

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May 2021 P1H | Question 20
Rounding, Estimation & Bounds

Topic check. Rounding, Estimation & Bounds is correct.

Method

$$P = \frac{t - w}{y}$$

For the upper bound of P, make the numerator as large as possible and the denominator as small as possible.
$t = 9.75, \quad w = 5.85, \quad y = 2.5$

$$P = \frac{9.75 - 5.85}{2.5}$$

$$P = \frac{3.90}{2.5} = 1.56$$

FINAL ANSWER
1.56

CLASSIFIED PROBLEM Paper: P1H | Session: Nov 2024 | Q#: 21 | Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Source: Nov 2024 P1H

21 $T = \dfrac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.
Show your working clearly.

....

(Total for Question 21 is 3 marks)

WORKED SOLUTION

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Nov 2024 P1H | Question 21

Rounding, Estimation & Bounds

Topic check. Rounding, Estimation & Bounds is correct.

Method

$$T = \frac{x^2 + y^2}{w}$$

For the upper bound of T, use the upper bounds of x and y, and the lower bound of w.
$x = 28.45, \quad y = 17.5, \quad w = 87.5$

$$T = \frac{28.45^2 + 17.5^2}{87.5}$$

$$T = 12.7503...$$

FINAL ANSWER

12.8 to 3 significant figures.

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2025 | Q#: 24 | Marks: 5

TOPIC: **Rounding, Estimation & Bounds**

Source: Nov 2025 P2H

24 $D = \dfrac{n}{p - q}$

$n = 10.3$ correct to 1 decimal place
$p = 7.24$ correct to 2 decimal places
$q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

....

(Total for Question 24 is 5 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Nov 2025 P2H | Question 24
Rounding, Estimation & Bounds

Topic check. Rounding, Estimation & Bounds is correct.

Method

$$D = \frac{n}{p-q}$$

Bounds:
$10.25 \leq n < 10.35$
$7.235 \leq p < 7.245$
$4.385 \leq q < 4.395$
Lower bound:

$$\frac{10.25}{7.245 - 4.385} = 3.5839...$$

Upper bound:

$$\frac{10.35}{7.235 - 4.395} = 3.6443...$$

So $3.5839... \leq D < 3.6443...$. Both bounds round to 3.6 to 1 decimal place.

FINAL ANSWER
$D = 3.6$ to a suitable degree of accuracy.

Topic 3

Surds

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2020 | Q#: 20 | Marks: 3

TOPIC: **Surds**

Source: Jan 2020 P2H

20 The area of a rectangle is 18 cm^2

The length of the rectangle is $\left(\sqrt{7}+1\right)$ cm.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.
Give your answer in the form $a\sqrt{b}+c$ where a, b and c are integers.

.... cm

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 20
Surds

Topic check. Corrected from Right-Angled Triangles to Surds.

Method

Area of rectangle:

$$18 \text{ cm}^2$$

Length:

$$\sqrt{7}+1$$

Width:

$$\frac{18}{\sqrt{7}+1}$$

Rationalise the denominator:

$$\begin{aligned}
&\frac{18}{\sqrt{7}+1} \times \frac{\sqrt{7}-1}{\sqrt{7}-1} \\
&= \frac{18(\sqrt{7}-1)}{7-1} \\
&= \frac{18(\sqrt{7}-1)}{6} \\
&= 3(\sqrt{7}-1) \\
&= 3\sqrt{7}-3
\end{aligned}$$

FINAL ANSWER

$\boxed{3\sqrt{7}-3 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jun 2022 | Q#: 21 | Marks: 4

Topic: **Surds**

Source: Jun 2022 P1HR

21 Express $\dfrac{3+\sqrt{8}}{\left(\sqrt{2}-1\right)^2}$ in the form $p+\sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

....

(Total for Question 21 is 4 marks)

WORKED SOLUTION

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Jun 2022 P1HR | Question 21
Surds

Topic check. Surds is correct.

Method

$$\frac{3+\sqrt{8}}{(\sqrt{2}-1)^2}$$

First simplify:

$$\sqrt{8} = 2\sqrt{2}$$

and

$$(\sqrt{2}-1)^2 = 2 - 2\sqrt{2} + 1 = 3 - 2\sqrt{2}$$

So

$$\frac{3+2\sqrt{2}}{3-2\sqrt{2}}$$

Rationalise the denominator:

$$\frac{3+2\sqrt{2}}{3-2\sqrt{2}} \times \frac{3+2\sqrt{2}}{3+2\sqrt{2}} = \frac{(3+2\sqrt{2})^2}{9-8}$$

$$(3+2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$$

FINAL ANSWER
$17 + 12\sqrt{2}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2023 | Q#: 23 | Marks: 4

TOPIC: **Surds**

Source: Jun 2023 P2H

23 The diagram shows a cuboid with a square cross section.

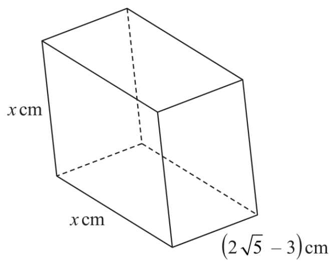

Diagram **NOT** accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\,\text{cm}^3$

Without using a calculator, find the value of x
Give your answer in the form $a + \sqrt{b}$ where a and b are integers.
Show your working clearly.

$x =$ ....

(Total for Question 23 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 23

Surds

Topic check. Corrected from Right-Angled Triangles to Surds.

Method

The cuboid has square cross section, so its volume is

$$x^2(2\sqrt{5}-3)$$

Given

$$x^2(2\sqrt{5}-3) = 13+6\sqrt{5}$$

So

$$x^2 = \frac{13+6\sqrt{5}}{2\sqrt{5}-3}$$

Rationalise the denominator:

$$x^2 = \frac{13+6\sqrt{5}}{2\sqrt{5}-3} \cdot \frac{2\sqrt{5}+3}{2\sqrt{5}+3}$$

$$x^2 = \frac{(13+6\sqrt{5})(2\sqrt{5}+3)}{20-9}$$

$$x^2 = \frac{99+44\sqrt{5}}{11}$$

$$x^2 = 9+4\sqrt{5}$$

Now

$$(2+\sqrt{5})^2 = 4+4\sqrt{5}+5 = 9+4\sqrt{5}$$

Therefore

$$x = 2+\sqrt{5}$$

FINAL ANSWER

$\boxed{2+\sqrt{5}}$

Topic 4

Algebraic Roots & Indices

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2020 | Q#: 21 | Marks: 7

TOPIC: **Algebraic Roots & Indices**

Source: Jan 2020 P2HR

21 (a) Simplify fully $\dfrac{10x^2 + 23x + 12}{4x^2 - 9}$

....

(3)

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

(b) Find an expression for n in terms of y.
Show clear algebraic working and simplify your expression.

....

(4)

(Total for Question 21 is 7 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jan 2020 P2HR | Question 21
Algebraic Roots & Indices

Topic check. Correct.

Method
(a)

$$\frac{10x^2+23x+12}{4x^2-9} = \frac{(5x+4)(2x+3)}{(2x-3)(2x+3)}$$
$$= \frac{5x+4}{2x-3}$$

(b)

$$2^{2y} \times 2^{3y+2} = 2^{5y+2}$$

Also,

$$\frac{8^{5y}}{4^n} = \frac{(2^3)^{5y}}{(2^2)^n} = 2^{15y-2n}$$

Equate powers:

$$5y+2 = 15y-2n$$

$$2n = 10y-2$$

$$n = 5y-1$$

FINAL ANSWER
(a) $\frac{5x+4}{2x-3}$, (b) $n = 5y-1$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2022 | Q#: 24 | Marks: 3

TOPIC: **Algebraic Roots & Indices**

Source: Jun 2022 P2H

24

$$\frac{18 \times \left(\sqrt{27}\right)^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

$x =$ ....

(Total for Question 24 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2H | Question 24

Algebraic Roots & Indices

Topic check. Correct.

Method

$$\frac{18(\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Use powers of 3:

$$18 \div 6 = 3$$

$$\sqrt{27} = (3^3)^{1/2} = 3^{3/2}$$

$$(\sqrt{27})^{4n+6} = 3^{6n+9}$$

$$9^{2n+8} = (3^2)^{2n+8} = 3^{4n+16}$$

So

$$\frac{3 \cdot 3^{6n+9}}{3^{4n+16}} = 3^{1+6n+9-4n-16} = 3^{2n-6}$$

Therefore,

$$x = 2n - 6$$

FINAL ANSWER

$x = 2n - 6$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2024 | Q#: 23 | Marks: 3

TOPIC: **Algebraic Roots & Indices**

Source: Jun 2024 P2H

23 Simplify $\dfrac{30 \times 25^{2x+7}}{\sqrt{180} \times \left(\sqrt{5}\right)^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

....

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 23

Algebraic Roots & Indices

Topic check. Correct.

Method

$$\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$$

$$\frac{30}{\sqrt{180}} = \frac{30}{6\sqrt{5}} = \sqrt{5} = 5^{1/2}$$

$$25^{2x+7} = (5^2)^{2x+7} = 5^{4x+14}$$

$$(\sqrt{5})^{4x+9} = 5^{2x+\frac{9}{2}}$$

So

$$5^{1/2} \times 5^{4x+14} \div 5^{2x+\frac{9}{2}} = 5^{\frac{1}{2}+4x+14-2x-\frac{9}{2}}$$

$$= 5^{2x+10}$$

FINAL ANSWER

5^{2x+10}, so $w = 2x + 10$

CLASSIFIED PROBLEM

Paper: P1HR | Session: May 2023 | Q#: 24 | Marks: 5

TOPIC: **Algebraic Roots & Indices**

Source: May 2023 P1HR

24 Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n
State any restrictions on n

....

(Total for Question 24 is 5 marks)

WORKED SOLUTION

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May 2023 P1HR | Question 24
Algebraic Roots & Indices

Topic check. Correct.

Method

$$2^n = 2^{x^2} \times 16^x \times 8$$

$$16^x = (2^4)^x = 2^{4x}$$

$$8 = 2^3$$

So

$$2^n = 2^{x^2+4x+3}$$

$$n = x^2 + 4x + 3$$

Complete the square:

$$n = (x+2)^2 - 1$$

$$(x+2)^2 = n + 1$$

Since $x > 0$, take the positive root:

$$x + 2 = \sqrt{n+1}$$

$$x = \sqrt{n+1} - 2$$

For $x > 0$,

$$\sqrt{n+1} > 2$$

$$n > 3$$

FINAL ANSWER
$x = \sqrt{n+1} - 2$, with $n > 3$

CLASSIFIED PROBLEM

Paper: P1HR | Session: MayNov 2020 | Q#: 21 | Marks: 5

Topic: **Algebraic Roots & Indices**

Source: May-Nov 2020 P1HR

21 Given that $M = \dfrac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

....

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | **Question 21**

Algebraic Roots & Indices

Topic check. Correct.

Method

$$M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$$

Write everything in powers of 2 and 3:

$$18^{4n} = (2 \cdot 3^2)^{4n} = 2^{4n}3^{8n}$$

$$2^{3(n^2-6n)} = 2^{3n^2-18n}$$

$$3^{2(1-4n)} = 3^{2-8n}$$

$$12^2 = (2^2 \cdot 3)^2 = 2^4 3^2$$

So

$$M = 2^{4n+3n^2-18n-4} \times 3^{8n+2-8n-2}$$

$$M = 2^{3n^2-14n-4}$$

Given $M = 2$,

$$2^{3n^2-14n-4} = 2^1$$

$$3n^2 - 14n - 4 = 1$$

$$3n^2 - 14n - 5 = 0$$

$$(3n+1)(n-5) = 0$$

FINAL ANSWER

$n = 5$ or $n = -\frac{1}{3}$

Topic 5

Completing the Square

CLASSIFIED PROBLEM Paper: 4WM1H Session: May2025 Q#: 25 Marks: 5

TOPIC: **Completing the Square**

Source: May 2025 4WM1H

25 (a) Express $8 - 12x - 4x^2$ in the form $a - b(x + c)^2$
where a, b and c are constants.

....

(3)

(b) Hence solve $8 - 12x - 4x^2 = 0$
Show your working clearly.
Give your solutions in surd form.

....

(2)

(Total for Question 25 is 5 marks)

WORKED SOLUTION

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May 2025 4WM1H | Question 25

Completing the Square

Topic check. Was tagged Surds; actually Completing the Square.

Solution

$$8x^2 - 12x - 4 = 8\left(x^2 - \frac{3}{2}x\right) - 4$$

Complete the square.

$$x^2 - \frac{3}{2}x = \left(x - \frac{3}{4}\right)^2 - \frac{9}{16}$$

So

$$8x^2 - 12x - 4 = 8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$$

Now solve

$$8x^2 - 12x - 4 = 0$$

$$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2} = 0$$

$$\left(x - \frac{3}{4}\right)^2 = \frac{17}{16}$$

$$x - \frac{3}{4} = \pm\frac{\sqrt{17}}{4}$$

$$x = \frac{3 \pm \sqrt{17}}{4}$$

FINAL ANSWER

$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$, and $x = \frac{3\pm\sqrt{17}}{4}$.

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 22 | Marks: 2

TOPIC: **Completing the Square**

Source: Nov 2025 4WM1H

22 $9x^2 - 12x + q$ can be written in the form $(px - r)^2$ where p, q and r are integers.

Find the value of q

$q = $ ....

(Total for Question 22 is 2 marks)

WORKED SOLUTION

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Nov 2025 4WM1H | Question 22

Completing the Square

Solution

$$9x^2 - 12x + q = (3x - 2)^2$$

$$(3x - 2)^2 = 9x^2 - 12x + 4$$

FINAL ANSWER

$q = 4.$

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 24 | Marks: 3

TOPIC: **Completing the Square**

Source: Nov 2025 4WM1H

24 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a, b and c are integers.

....

(Total for Question 24 is 3 marks)

WORKED SOLUTION

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Nov 2025 4WM1H | Question 24

Completing the Square

Solution

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5((x-2)^2 - 4) + 23$$

$$= 5(x-2)^2 + 3$$

FINAL ANSWER

$5(x-2)^2 + 3$.

Topic 6

Algebraic Fractions

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: May2025 | Q#: 20 | Marks: 3

TOPIC: **Algebraic Fractions**

Source: May 2025 4WM1H

20 Given that $k = m + n$ and $m = \dfrac{3}{4n}$

express $\dfrac{7k}{5 - m}$ in the form $\dfrac{a + bn^2}{cn - d}$ where a, b and c are integers and d is prime.

....

(Total for Question 20 is 3 marks)

WORKED SOLUTION

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May 2025 4WM1H | Question 20
Algebraic Fractions

Topic check. Was tagged Algebra Toolkit; actually Algebraic Fractions.

Solution

$$k = m + n, \qquad m = \frac{3}{4n}$$

So

$$k = \frac{3}{4n} + n = \frac{3 + 4n^2}{4n}$$

Then

$$7k = \frac{21 + 28n^2}{4n}$$

and

$$5 - m = 5 - \frac{3}{4n} = \frac{20n - 3}{4n}$$

Therefore

$$\frac{7k}{5 - m} = \frac{\frac{21+28n^2}{4n}}{\frac{20n-3}{4n}}$$
$$= \frac{21 + 28n^2}{20n - 3}$$

FINAL ANSWER
$\frac{21+28n^2}{20n-3}$

CLASSIFIED PROBLEM Paper: 4WM1HR Session: May2025 Q#: 24 Marks: 4

TOPIC: **Algebraic Fractions**

Source: May 2025 4WM1HR

24 Write $(2x-1)+\dfrac{2x^2-5x-12}{x^2-16}\times\dfrac{x^2+4x}{6x^2+11x+3}$ as a single fraction in its simplest form.

Show clear algebraic working.

....

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 24

Algebraic Fractions

Solution

Factor the fractional part first:

$$\frac{2x^2-5x-12}{x^2-16}\times\frac{x^2+4x}{6x^2+11x+3}$$

$$=\frac{(2x+3)(x-4)}{(x-4)(x+4)}\times\frac{x(x+4)}{(3x+1)(2x+3)}$$

$$=\frac{x}{3x+1}$$

So

$$(2x-1)+\frac{x}{3x+1}=\frac{(2x-1)(3x+1)+x}{3x+1}$$

$$=\frac{6x^2-1}{3x+1}$$

FINAL ANSWER

$\dfrac{6x^2-1}{3x+1}$.

CLASSIFIED PROBLEM Paper: 4WM1H Session: Nov2025 Q#: 25 Marks: 3

TOPIC: **Algebraic Fractions**

Source: Nov 2025 4WM1H

25 $a = \dfrac{2x+5}{1-x}$ $\quad x = \dfrac{5-2y}{3y}$

Write a in the form $\dfrac{m+ny}{p(y-1)}$ where m, n and p are integers.

Show your working clearly.

$a =$ ....

(Total for Question 25 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 25

Algebraic Fractions

Topic check. Corrected from Ratio Toolkit to Algebraic Fractions.

Method

$$a = \frac{2x+5}{1-x}, \quad x = \frac{5-2y}{3y}$$

Substitute for x.
Numerator:

$$2x+5 = 2\left(\frac{5-2y}{3y}\right)+5$$

$$= \frac{10-4y+15y}{3y} = \frac{10+11y}{3y}$$

Denominator:

$$1-x = 1-\frac{5-2y}{3y}$$

$$= \frac{3y-5+2y}{3y} = \frac{5y-5}{3y} = \frac{5(y-1)}{3y}$$

Therefore

$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10+11y}{5(y-1)}$$

FINAL ANSWER

$$\frac{10+11y}{5(y-1)}$$

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 28 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Nov 2025 4WM1H

28 Solve $\dfrac{6x^3 + 5x^2 + x}{4x^2 - 1} \div \dfrac{15x^2 - x - 2}{8x - 4} = 2x$

Show clear algebraic working.

$x = $ ....

(Total for Question 28 is 4 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Nov 2025 4WM1H | Question 28
Algebraic Fractions

Solution
Factor each expression:

$$\frac{x(3x+1)(2x+1)}{(2x-1)(2x+1)} \div \frac{(5x-2)(3x+1)}{4(2x-1)} = 2x$$

$$\frac{x(3x+1)}{2x-1} \times \frac{4(2x-1)}{(5x-2)(3x+1)} = 2x$$

$$\frac{4x}{5x-2} = 2x$$

$$4x = 2x(5x-2)$$

$$10x^2 - 8x = 0$$

$$2x(5x-4) = 0$$

The original fractions do not allow $x = 0$, so

$$x = \frac{4}{5}$$

FINAL ANSWER
$x = \frac{4}{5}$.

Topic 7

Completing the Square

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2020 | Q#: 23 | Marks: 3

TOPIC: **Completing the Square**

Source: Jan 2020 P2HR

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

....

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 23
Completing the Square

Topic check. Correct.

Method

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

$$= -2(x^2 + 6x) + 7$$

$$= -2\left((x+3)^2 - 9\right) + 7$$

$$= -2(x+3)^2 + 18 + 7$$

$$= 25 - 2(x+3)^2$$

FINAL ANSWER
$25 - 2(x+3)^2$

CLASSIFIED PROBLEM

Paper: P1H | Session: Jan 2022 | Q#: 20 | Marks: 4

TOPIC: **Completing the Square**

Source: Jan 2022 P1H

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

....

(3)

C is the curve with equation $y = 7 + 12x - 3x^2$

The point A is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of A

(.................... , ...)

(1)

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 20

Completing the Square

Topic check. Correct.

Method

(a)

$$7 + 12x - 3x^2 = -3x^2 + 12x + 7$$

$$= -3(x^2 - 4x) + 7$$

$$= -3\left((x-2)^2 - 4\right) + 7$$

$$= 19 - 3(x-2)^2$$

This is the same as

$$19 - 3(x + (-2))^2$$

(b) The maximum point is read from the completed square form.

$$y = 19 - 3(x-2)^2$$

The maximum occurs when $x = 2$, giving $y = 19$.

FINAL ANSWER

(a) $19 - 3(x-2)^2$, (b) $A = (2, 19)$

CLASSIFIED PROBLEM

Paper: P1H **Session: Jun 2022** **Q#: 24** **Marks: 4**

TOPIC: **Completing the Square**

Source: Jun 2022 P1H

24 Express each of a, b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$a =$ ....

$b =$ ....

$c =$ ....

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 24

Completing the Square

Topic check. Correct.

Method

$$q + 12x - qx^2 = -q\left(x^2 - \frac{12}{q}x\right) + q$$

Complete the square:

$$x^2 - \frac{12}{q}x = \left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}$$

So

$$q + 12x - qx^2 = -q\left[\left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}\right] + q$$

$$= q + \frac{36}{q} - q\left(x - \frac{6}{q}\right)^2$$

Compare with

$$a - b(x - c)^2$$

FINAL ANSWER

$a = q + \dfrac{36}{q}$, $b = q$, $c = \dfrac{6}{q}$

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jun 2022 | Q#: 23 | Marks: 5

TOPIC: **Completing the Square**

Source: Jun 2022 P1HR

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

....

(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(.................... , ...)

(2)

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 23
Completing the Square

Topic check. Correct.

Method

(a)

$$2x^2 - 12x + 3 = 2(x^2 - 6x) + 3$$

$$= 2\left((x-3)^2 - 9\right) + 3$$

$$= 2(x-3)^2 - 18 + 3$$

$$= 2(x-3)^2 - 15$$

(b)

$$y = 2(x+4)^2 - 12(x+4) + 3$$

Let $u = x + 4$. From part (a),

$$2u^2 - 12u + 3 = 2(u-3)^2 - 15$$

So

$$y = 2((x+4) - 3)^2 - 15$$

$$y = 2(x+1)^2 - 15$$

The minimum point is when $x = -1$, giving $y = -15$.

FINAL ANSWER
(a) $2(x-3)^2 - 15$, (b) $M = (-1, -15)$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2025 | Q#: 25 | Marks: 6

TOPIC: **Completing the Square**

Source: Jun 2025 P2H

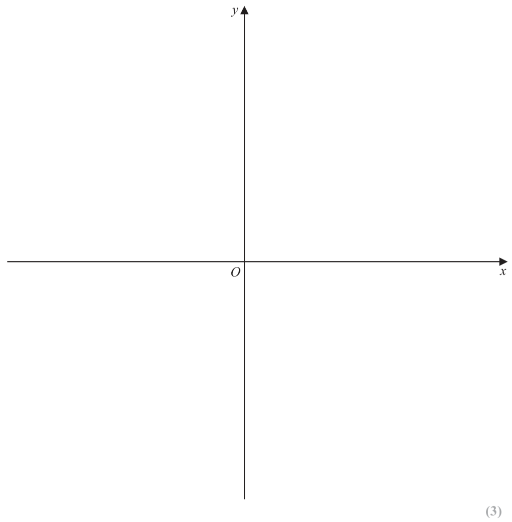
25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a, b and c are integers.

....

(3)

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y-axis.



(3)

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 25

Completing the Square

Topic check. Correct.

Method

(a)

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

$$= -6(x^2 - 4x) + 28$$

$$= -6\left((x-2)^2 - 4\right) + 28$$

$$= 52 - 6(x-2)^2$$

(b)

$$y = 52 - 6(x-2)^2$$

The graph is a downward-opening parabola.
The turning point is

$$(2, 52)$$

The point of intersection with the y-axis is found by putting $x = 0$:

$$y = 28$$

So the y-intercept is

$$(0, 28)$$

FINAL ANSWER

(a) $52 - 6(x-2)^2$. For the sketch, show a downward parabola with turning point $(2, 52)$ and y-intercept $(0, 28)$.

CLASSIFIED PROBLEM

Paper: P2H | Session: May 2021 | Q#: 22 | Marks: 4

TOPIC: **Completing the Square**

Source: May 2021 P2H

22 The curve **S** has equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = x^2$
The curve **T** has equation $y = \mathrm{g}(x)$ where $\mathrm{g}(x) = 2x^2 - 12x + 13$

By writing $\mathrm{g}(x)$ in the form $a(x - b)^2 - c$, where a, b and c are constants, describe fully a series of transformations that map the curve **S** onto the curve **T**.

....

....

....

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P2H | Question 22

Completing the Square

Topic check. Correct.

Method

$$g(x) = 2x^2 - 12x + 13$$

$$= 2(x^2 - 6x) + 13$$

$$= 2\left((x-3)^2 - 9\right) + 13$$

$$= 2(x-3)^2 - 18 + 13$$

$$= 2(x-3)^2 - 5$$

Starting with $S : y = x^2$:

- Translate 3 units to the right to get $y = (x-3)^2$.
- Stretch parallel to the y-axis by scale factor 2 to get $y = 2(x-3)^2$.
- Translate 5 units down to get $y = 2(x-3)^2 - 5$.

FINAL ANSWER

$g(x) = 2(x-3)^2 - 5$. Transformations: right 3, stretch parallel to the y-axis by scale factor 2, then down 5.

CLASSIFIED PROBLEM

Paper: P1H **Session: May 2024** **Q#: 25** **Marks: 4**

TOPIC: **Completing the Square**

Source: May 2024 P1H

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a, b and c are constants.

$f(x) =$ ....

(Total for Question 25 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 25

Completing the Square

Topic check. Correct.

Method

$$
\begin{aligned}
f(x) &= 17 - 3x^2 + 12x \\
&= -3x^2 + 12x + 17 \\
&= -3(x^2 - 4x) + 17 \\
&= -3\left((x-2)^2 - 4\right) + 17 \\
&= -3(x-2)^2 + 12 + 17 \\
&= 29 - 3(x-2)^2
\end{aligned}
$$

FINAL ANSWER

$f(x) = 29 - 3(x-2)^2$

CLASSIFIED PROBLEM

Paper: P1HR **Session: May 2024** **Q#: 20** **Marks: 5**

TOPIC: **Completing the Square**

Source: May 2024 P1HR

20 (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a, b and c are numbers to be found.

....
(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

(b) Find the coordinates of P

(.......................... , ...)
(2)

(Total for Question 20 is 5 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

May 2024 P1HR | Question 20
Completing the Square

Topic check. Moved from Coordinate Geometry to Completing the Square.

Method

For part (a),

$$2x^2 - 11x + 9$$

Factor out 2 from the quadratic terms:

$$2\left(x^2 - \frac{11}{2}x\right) + 9$$

Complete the square:

$$2\left[\left(x - \frac{11}{4}\right)^2 - \frac{121}{16}\right] + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + \frac{72}{8}$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

So

$$2x^2 - 11x + 9 = 2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

For part (b),

$$y = 2(x-3)^2 - 11(x-3) + 9$$

Let

$$u = x - 3$$

From part (a), the minimum occurs when

$$u = \frac{11}{4}$$

So

$$x - 3 = \frac{11}{4}$$

$$x = \frac{23}{4}$$

The minimum value of y is

$$-\frac{49}{8}$$

FINAL ANSWER

$2\left(x-\frac{11}{4}\right)^2-\frac{49}{8}$, $P=\left(\frac{23}{4},-\frac{49}{8}\right)$

CLASSIFIED PROBLEM Paper: P1H Session: Nov 2025 Q#: 21 Marks: 3

TOPIC: **Completing the Square**

Source: Nov 2025 P1H

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a, b and c are integers.

....

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 21

Completing the Square

Topic check. Correct.

Method

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5\left((x-2)^2 - 4\right) + 23$$

$$= 5(x-2)^2 - 20 + 23$$

$$= 5(x-2)^2 + 3$$

FINAL ANSWER

$5(x-2)^2 + 3$

Topic 8

Algebraic Fractions

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2020 | Q#: 24 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jan 2020 P2H

24 Express

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

....

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 24
Algebraic Fractions

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$\left(\frac{4}{2x-5} - \frac{3}{2x-3}\right) \div \frac{9x-4x^3}{6x^2-17x+5}$$

First simplify the bracket:

$$\begin{aligned}\frac{4}{2x-5} - \frac{3}{2x-3} &= \frac{4(2x-3)-3(2x-5)}{(2x-5)(2x-3)} \\ &= \frac{8x-12-6x+15}{(2x-5)(2x-3)} \\ &= \frac{2x+3}{(2x-5)(2x-3)}\end{aligned}$$

Now factor the second fraction:

$$9x-4x^3 = -x(2x-3)(2x+3)$$

$$6x^2-17x+5 = (2x-5)(3x-1)$$

Dividing by a fraction means multiplying by its reciprocal:

$$\frac{2x+3}{(2x-5)(2x-3)} \times \frac{(2x-5)(3x-1)}{-x(2x-3)(2x+3)}$$

Cancel common factors:

$$\begin{aligned}&= -\frac{3x-1}{x(2x-3)^2} \\ &= \frac{1-3x}{x(2x-3)^2}\end{aligned}$$

FINAL ANSWER

$$\boxed{\frac{1-3x}{x(2x-3)^2}}$$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2021 | Q#: 21 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jan 2021 P2H

21 Write $\dfrac{25x^2 - 64}{5x^2 - 13x - 6} \times \dfrac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

....

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2H | Question 21
Algebraic Fractions

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$\frac{25x^2-64}{5x^2-13x-6} \times \frac{x^2-8x+15}{5x+8} - (x-7)$$

Factor:

$$25x^2-64 = (5x-8)(5x+8)$$

$$5x^2-13x-6 = (5x+2)(x-3)$$

$$x^2-8x+15 = (x-3)(x-5)$$

So

$$\frac{(5x-8)(5x+8)}{(5x+2)(x-3)} \times \frac{(x-3)(x-5)}{5x+8} - (x-7)$$

Cancel common factors:

$$\frac{(5x-8)(x-5)}{5x+2} - (x-7)$$

Use the common denominator $5x+2$:

$$\frac{(5x-8)(x-5)-(x-7)(5x+2)}{5x+2}$$

Expand:

$$(5x-8)(x-5) = 5x^2-33x+40$$

$$(x-7)(5x+2) = 5x^2-33x-14$$

So the numerator is

$$5x^2-33x+40-(5x^2-33x-14) = 54$$

FINAL ANSWER

$$\boxed{\frac{54}{5x+2}}$$

CLASSIFIED PROBLEM

Paper: P2HR **Session: Jun 2022** **Q#: 23** **Marks: 3**

TOPIC: **Algebraic Fractions**

Source: Jun 2022 P2HR

23 Express $\left(\dfrac{20}{x^2-36} - \dfrac{2}{x-6}\right) \times \dfrac{1}{4-x}$ as a single fraction in its simplest form.

....

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 23

Algebraic Fractions

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$\left(\frac{20}{x^2-36}-\frac{2}{x-6}\right)\times\frac{1}{4-x}$$

Factor:

$$x^2-36=(x-6)(x+6)$$

So

$$\frac{20}{(x-6)(x+6)}-\frac{2}{x-6}$$

Use the common denominator $(x-6)(x+6)$:

$$\frac{20-2(x+6)}{(x-6)(x+6)}$$

$$=\frac{20-2x-12}{(x-6)(x+6)}$$

$$=\frac{8-2x}{(x-6)(x+6)}$$

$$=\frac{-2(x-4)}{(x-6)(x+6)}$$

Now multiply by

$$\frac{1}{4-x}=-\frac{1}{x-4}$$

$$\frac{-2(x-4)}{(x-6)(x+6)}\times\left(-\frac{1}{x-4}\right)$$

Cancel $x-4$:

$$\frac{2}{(x-6)(x+6)}$$

$$=\frac{2}{x^2-36}$$

FINAL ANSWER

$$\boxed{\frac{2}{x^2-36}}$$

CLASSIFIED PROBLEM Paper: P2H | Session: Jun 2024 | Q#: 20 | Marks: 3

TOPIC: **Algebraic Fractions**

Source: Jun 2024 P2H

20 Given that $k = x - y$ and $x = \dfrac{1}{4y}$

express $\dfrac{5k}{x+2}$ in the form $\dfrac{a - by^2}{c + dy}$ where a, b, c and d are integers.

....

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 20

Algebraic Fractions

Topic check. Corrected from Direct & Inverse Proportion to Algebraic Fractions.

Method

$$k = x - y, \quad x = \frac{1}{4y}$$

So

$$k = \frac{1}{4y} - y = \frac{1 - 4y^2}{4y}$$

Also

$$x + 2 = \frac{1}{4y} + 2 = \frac{1 + 8y}{4y}$$

Therefore

$$\frac{5k}{x+2} = \frac{5\left(\frac{1-4y^2}{4y}\right)}{\frac{1+8y}{4y}}$$

Cancel $4y$:

$$\frac{5(1 - 4y^2)}{1 + 8y}$$

$$= \frac{5 - 20y^2}{1 + 8y}$$

FINAL ANSWER

$$\frac{5 - 20y^2}{1 + 8y}$$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2024 | Q#: 26 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jun 2024 P2HR

26 Write $4 - \left[(3x - 5) \div \dfrac{3x^2 + x - 10}{4x - 1}\right]$ as a single fraction in its simplest form.

....

(Total for Question 26 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 26

Algebraic Fractions

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$4 - \left[(3x-5) \div \frac{3x^2+x-10}{4x-1}\right]$$

Factor the quadratic:

$$3x^2 + x - 10 = (3x-5)(x+2)$$

So the expression inside the square brackets is

$$(3x-5) \div \frac{(3x-5)(x+2)}{4x-1}$$

Dividing by a fraction means multiplying by its reciprocal:

$$(3x-5) \times \frac{4x-1}{(3x-5)(x+2)}$$

Cancel $3x-5$:

$$\frac{4x-1}{x+2}$$

Therefore the full expression is

$$4 - \frac{4x-1}{x+2}$$

Use the common denominator $x+2$:

$$\frac{4(x+2)-(4x-1)}{x+2}$$

$$= \frac{4x+8-4x+1}{x+2}$$

$$= \frac{9}{x+2}$$

FINAL ANSWER

$$\boxed{\frac{9}{x+2}}$$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2025 | Q#: 23 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jun 2025 P2H

23 $\dfrac{4x^2-4x-120}{5x^2-180} \div \dfrac{x^2+5x}{10x^2+60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$p =$ ....

(Total for Question 23 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 23

Algebraic Fractions

Topic check. Corrected from Ratio Toolkit to Algebraic Fractions.

Method

$$\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x}$$

Factorise:

$$4x^2 - 4x - 120 = 4(x-6)(x+5)$$

$$5x^2 - 180 = 5(x-6)(x+6)$$

$$x^2 + 5x = x(x+5)$$

$$10x^2 + 60x = 10x(x+6)$$

So

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)}$$

$$= \frac{4(x+5)}{5(x+6)} \times \frac{10(x+6)}{x+5} = 8$$

FINAL ANSWER

$p = 8$

CLASSIFIED PROBLEM

Paper: P1H | Session: May 2021 | Q#: 21 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: May 2021 P1H

21 Given that $x = \dfrac{5}{9y+5}$ and that $y = \dfrac{5}{5a-2}$

find an expression for x in terms of a.
Give your expression as a single fraction in its simplest form.

....

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 21

Algebraic Fractions

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

Given

$$x = \frac{5}{9y+5}$$

and

$$y = \frac{5}{5a-2}$$

Substitute $y = \frac{5}{5a-2}$ into the expression for x:

$$x = \frac{5}{9\left(\frac{5}{5a-2}\right)+5}$$

$$x = \frac{5}{\frac{45}{5a-2}+5}$$

Write the denominator as a single fraction:

$$\frac{45}{5a-2}+5 = \frac{45+5(5a-2)}{5a-2}$$

$$= \frac{45+25a-10}{5a-2}$$

$$= \frac{25a+35}{5a-2}$$

So

$$x = 5 \div \frac{25a+35}{5a-2}$$

$$x = 5 \times \frac{5a-2}{25a+35}$$

$$x = \frac{5(5a-2)}{5(5a+7)}$$

$$x = \frac{5a-2}{5a+7}$$

FINAL ANSWER

$$\boxed{\frac{5a-2}{5a+7}}$$

CLASSIFIED PROBLEM

Paper: P1HR | Session: May 2023 | Q#: 23 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: May 2023 P1HR

23 Simplify $(x^2 - 4) \div \left(\dfrac{4x^2 - 7x - 2}{x}\right) - 2x$

Give your answer in the form $\dfrac{ax^2}{bx + c}$ where a, b and c are integers.

....

(Total for Question 23 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 23
Algebraic Fractions

Topic check. Corrected from Surds to Algebraic Fractions.

Method

$$(x^2 - 4) \div \left(\frac{4x^2 - 7x - 2}{x}\right) - 2x$$

Change division into multiplication by the reciprocal:

$$(x^2 - 4) \times \frac{x}{4x^2 - 7x - 2} - 2x$$

Factorise:

$$x^2 - 4 = (x-2)(x+2)$$

$$4x^2 - 7x - 2 = (4x+1)(x-2)$$

So

$$\frac{x(x-2)(x+2)}{(4x+1)(x-2)} - 2x = \frac{x(x+2)}{4x+1} - 2x$$

$$\frac{x(x+2) - 2x(4x+1)}{4x+1} = \frac{x^2 + 2x - 8x^2 - 2x}{4x+1} = \frac{-7x^2}{4x+1}$$

FINAL ANSWER

$$\frac{-7x^2}{4x+1}$$

CLASSIFIED PROBLEM Paper: P1H | Session: MayNov 2020 | Q#: 21 | Marks: 5

TOPIC: **Algebraic Fractions**

Source: May-Nov 2020 P1H

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

....

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 21

Algebraic Fractions

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

Factor:

$$9x^2 - 4 = (3x-2)(3x+2)$$

$$3x^2 - 13x - 10 = (3x+2)(x-5)$$

So the first product becomes

$$\frac{1}{3x-2} \times \frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$$

Cancel common factors:

$$\frac{1}{x-5}$$

Now subtract:

$$\frac{1}{x-5} - \frac{7}{x-1}$$

Use the common denominator $(x-5)(x-1)$:

$$\frac{x-1-7(x-5)}{(x-5)(x-1)}$$

$$= \frac{x-1-7x+35}{(x-5)(x-1)}$$

$$= \frac{34-6x}{(x-5)(x-1)}$$

FINAL ANSWER

$$\boxed{\frac{34-6x}{(x-5)(x-1)}}$$

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2023 | Q#: 24 | Marks: 5

TOPIC: **Algebraic Fractions**

Source: Nov 2023 P2H

24 Solve $\dfrac{45x^3 - 80x}{3x^2 + x - 4} \times \left(\dfrac{1}{3x-4} + \dfrac{1}{x}\right) = \dfrac{4(x+2)}{5x-8}$

Show clear algebraic working.

$x =$ ....

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | Question 24

Algebraic Fractions

Topic check. Corrected from Solving Quadratic Equations to Algebraic Fractions.

Method

$$\frac{45x^3 - 80x}{3x^2 + x - 4}\left(\frac{1}{3x-4} + \frac{1}{x}\right) = \frac{4(x+2)}{5x-8}$$

Factorise:

$$45x^3 - 80x = 5x(3x-4)(3x+4)$$

$$3x^2 + x - 4 = (3x+4)(x-1)$$

Also,

$$\frac{1}{3x-4} + \frac{1}{x} = \frac{x + 3x - 4}{x(3x-4)} = \frac{4(x-1)}{x(3x-4)}$$

So the left side becomes

$$\frac{5x(3x-4)(3x+4)}{(3x+4)(x-1)} \times \frac{4(x-1)}{x(3x-4)} = 20$$

Therefore,

$$20 = \frac{4(x+2)}{5x-8}$$

$$20(5x-8) = 4(x+2)$$

$$100x - 160 = 4x + 8$$

$$96x = 168$$

$$x = \frac{7}{4}$$

FINAL ANSWER

$x = \frac{7}{4}$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2025 | Q#: 23 | Marks: 3

TOPIC: **Algebraic Fractions**

Source: Nov 2025 P1H

23 $a = \dfrac{2x+5}{1-x} \qquad x = \dfrac{5-2y}{3y}$

Write a in the form $\dfrac{m+ny}{p(y-1)}$ where m, n and p are integers.

Show your working clearly.

$a =$ ....

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 23

Algebraic Fractions

Topic check. Corrected from Ratio Toolkit to Algebraic Fractions.

Method

$$a = \frac{2x+5}{1-x}, \quad x = \frac{5-2y}{3y}$$

Substitute for x.
Numerator:

$$2x + 5 = 2\left(\frac{5-2y}{3y}\right) + 5$$

$$= \frac{10-4y+15y}{3y} = \frac{10+11y}{3y}$$

Denominator:

$$1 - x = 1 - \frac{5-2y}{3y}$$

$$= \frac{3y-5+2y}{3y} = \frac{5y-5}{3y} = \frac{5(y-1)}{3y}$$

Therefore

$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10+11y}{5(y-1)}$$

FINAL ANSWER

$$\frac{10+11y}{5(y-1)}$$

Topic 9

Coordinate Geometry

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: May2025 | Q#: 24 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: May 2025 4WM1H

24 *PQRS* is a square.

PR is a diagonal of the square.

P is the point with coordinates (4, 7)
R is the point with coordinates (8, –5)

Find an equation of the straight line that passes through the points *Q* and *S*
Give your answer in the form $ay = bx + c$ where a, b and c are integers.

....

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 24

Coordinate Geometry

Topic check. Was tagged Algebra Toolkit; actually Linear Graphs.

Solution

The diagonals of a square bisect each other and are perpendicular.
Midpoint of $P(4,7)$ and $R(8,-5)$:

$$\left(\frac{4+8}{2}, \frac{7-5}{2}\right) = (6,1)$$

Gradient of PR:

$$\frac{-5-7}{8-4} = -3$$

So the perpendicular gradient is $\frac{1}{3}$.
The line through Q and S passes through $(6,1)$, so

$$y - 1 = \frac{1}{3}(x-6)$$

$$3y - 3 = x - 6$$

$$3y = x - 3$$

FINAL ANSWER

$3y = x - 3$

CLASSIFIED PROBLEM

Paper: 4WM1HR | Session: May2025 | Q#: 23 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: May 2025 4WM1HR

23 A circle C has centre $(3q, 4q)$ where q is an integer.

The straight line with equation $y = -\frac{5}{4}x + \frac{21}{4}$ is the tangent to C at the point $(p, p+3)$ where p is an integer.

Work out the radius of C
Give your answer in the form $\sqrt{a}$ where a is an integer.
Show your working clearly.

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 23

Coordinate Geometry

Solution

The tangent point lies on the tangent line, so

$$p + 3 = -\frac{5}{4}p + \frac{21}{4}$$

$$p = 1$$

The tangent gradient is $-\frac{5}{4}$, so the radius gradient is $\frac{4}{5}$.
Using centre $(3q, 4q)$ and tangent point $(1, 4)$,

$$\frac{4 - 4q}{1 - 3q} = \frac{4}{5}$$

$$q = 2$$

So the centre is $(6, 8)$. Therefore

$$r = \sqrt{(6-1)^2 + (8-4)^2} = \sqrt{41}$$

FINAL ANSWER

$\sqrt{41}$.

CLASSIFIED PROBLEM Paper: 4WM1H Session: Nov2025 Q#: 27 Marks: 5

TOPIC: **Coordinate Geometry**

Source: Nov 2025 4WM1H

27 $ABCD$ is a rhombus with $AB = BC = CD = DA$

A has coordinates $(16, -9)$

C has coordinates $(24, 15)$

Find an equation of the line BD
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.
Show your working clearly.

....

(Total for Question 27 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 27
Coordinate Geometry

Solution

Diagonals of a rhombus bisect each other at right angles.
Midpoint of AC:

$$\left(\frac{16+24}{2}, \frac{-9+15}{2}\right) = (20, 3)$$

Gradient of AC:

$$\frac{15-(-9)}{24-16} = 3$$

So gradient of BD is $-\frac{1}{3}$.

$$y - 3 = -\frac{1}{3}(x - 20)$$

$$x + 3y - 29 = 0$$

FINAL ANSWER

$x + 3y - 29 = 0$.

CLASSIFIED PROBLEM Paper: P1H Session: Jan 2020 Q#: 22 Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jan 2020 P1H

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$
J is the point with coordinates $(j, 15)$ where $j < 0$
K is the point with coordinates $(6, k)$

M is the midpoint of JK.
The gradient of HM is 2

Find the value of j and the value of k.

$j =$ ....

$k =$ ....

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1H | Question 22
Coordinate Geometry

Topic check. Correct.

Method

Since M is the midpoint of JK,

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2}\right)$$

The gradient of HM is 2, and $H = (-4, 1)$.

$$\frac{\frac{15+k}{2} - 1}{\frac{j+6}{2} + 4} = 2$$

$$\frac{k+13}{j+14} = 2$$

$$k = 2j + 15$$

Also,

$$JK = \sqrt{80}$$

So

$$(6-j)^2 + (k-15)^2 = 80$$

Substitute $k = 2j + 15$:

$$(6-j)^2 + (2j)^2 = 80$$

$$j^2 - 12j + 36 + 4j^2 = 80$$

$$5j^2 - 12j - 44 = 0$$

$$(5j+10)(j-4.4) = 0$$

Since $j < 0$,

$$j = -2$$

Then

$$k = 2(-2) + 15 = 11$$

FINAL ANSWER
$j = -2,\ k = 11$

CLASSIFIED PROBLEM

Paper: P2H **Session: Jan 2020** **Q#: 22** **Marks: 5**

TOPIC: **Coordinate Geometry**

Source: Jan 2020 P2H

22 The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B.

Find the coordinates of A and B.
Show clear algebraic working.

(.......................... , ...)

(.......................... , ...)

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 22
Coordinate Geometry

Topic check. Correct.

Method

The line is

$$y = x + 2$$

Substitute this into the circle equation:

$$x^2 + y^2 - 2y = 24$$

$$x^2 + (x+2)^2 - 2(x+2) = 24$$

$$x^2 + x^2 + 4x + 4 - 2x - 4 = 24$$

$$2x^2 + 2x - 24 = 0$$

$$x^2 + x - 12 = 0$$

$$(x+4)(x-3) = 0$$

So

$$x = -4 \quad \text{or} \quad x = 3$$

Using $y = x + 2$:

$$x = -4 \Rightarrow y = -2$$

$$x = 3 \Rightarrow y = 5$$

FINAL ANSWER
$(-4, -2)$ and $(3, 5)$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2020 | Q#: 24 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2020 P2HR

24 $\mathbf{L}_1$ and $\mathbf{L}_2$ are two straight lines.
The origin of the coordinate axes is O.

$\mathbf{L}_1$ has equation $5x + 10y = 8$
$\mathbf{L}_2$ is perpendicular to $\mathbf{L}_1$ and passes through the point with coordinates (8, 6)

$\mathbf{L}_2$ crosses the x-axis at the point A.
$\mathbf{L}_2$ intersects the straight line with equation $x = -3$ at the point B.

Find the area of triangle AOB.
Show your working clearly.

....

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 24

Coordinate Geometry

Topic check. Correct.

Method

For L_1,

$$5x + 10y = 8$$

$$10y = -5x + 8$$

$$y = -\frac{1}{2}x + \frac{4}{5}$$

So the gradient of L_1 is $-\frac{1}{2}$.
Since L_2 is perpendicular to L_1, the gradient of L_2 is 2.
Using the point $(8, 6)$:

$$y - 6 = 2(x - 8)$$

$$y = 2x - 10$$

At A, the line crosses the x-axis, so $y = 0$:

$$0 = 2x - 10$$

$$x = 5$$

So $A = (5, 0)$.
At B, $x = -3$:

$$y = 2(-3) - 10 = -16$$

So $B = (-3, -16)$.
Triangle AOB has base $OA = 5$ and perpendicular height 16.

$$\text{Area} = \frac{1}{2} \times 5 \times 16 = 40$$

FINAL ANSWER

40

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jan 2021 | Q#: 22 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2021 P1HR

22 ABC is an isosceles triangle with $AB = AC$.

B is the point with coordinates $(-1, 5)$
C is the point with coordinates $(2, 10)$
M is the midpoint of BC.

Find an equation of the line through the points A and M.
Give your answer in the form $py + qx = r$ where p, q and r are integers.

....

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1HR | Question 22
Coordinate Geometry

Topic check. Correct.

Method

Since $AB = AC$, point A lies on the perpendicular bisector of BC.
The midpoint of BC is

$$M = \left(\frac{-1+2}{2}, \frac{5+10}{2}\right)$$

$$M = \left(\frac{1}{2}, \frac{15}{2}\right)$$

The gradient of BC is

$$\frac{10-5}{2-(-1)} = \frac{5}{3}$$

So the gradient of the perpendicular bisector is

$$-\frac{3}{5}$$

Use the line through M:

$$y - \frac{15}{2} = -\frac{3}{5}\left(x - \frac{1}{2}\right)$$

Multiply by 10:

$$10y - 75 = -6x + 3$$

$$10y + 6x = 78$$

Divide by 2:

$$5y + 3x = 39$$

FINAL ANSWER
$5y + 3x = 39$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2022 | Q#: 22 | Marks: 5

Topic: **Coordinate Geometry**

Source: Jan 2022 P2HR

22 ABC is a triangle in which angle $ABC = 90°$

p and q are integers such that

the coordinates of A are $(p, 10)$
the coordinates of B are $(-1, -5)$
the coordinates of C are $(8, q)$

Given that the gradient of AC is $-\frac{6}{7}$

work out the value of p and the value of q

$p =$ ....

$q =$ ....

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2HR | Question 22

Coordinate Geometry

Topic check. Correct.

Method

The gradient of AC is $-\frac{6}{7}$.

$$\frac{q-10}{8-p} = -\frac{6}{7}$$

$$7(q-10) = -6(8-p)$$

$$7q - 70 = -48 + 6p$$

$$6p - 7q = -22$$

Since $\angle ABC = 90°$, lines AB and BC are perpendicular.

$$\text{gradient of } AB = \frac{10-(-5)}{p-(-1)} = \frac{15}{p+1}$$

$$\text{gradient of } BC = \frac{q-(-5)}{8-(-1)} = \frac{q+5}{9}$$

The product of perpendicular gradients is -1:

$$\frac{15}{p+1} \times \frac{q+5}{9} = -1$$

$$15(q+5) = -9(p+1)$$

$$5(q+5) = -3(p+1)$$

$$3p + 5q = -28$$

Now solve

$$6p - 7q = -22$$

$$3p + 5q = -28$$

Double the second equation:

$$6p + 10q = -56$$

Subtract:

$$17q = -34$$

$$q = -2$$

Then

$$3p + 5(-2) = -28$$

$$3p = -18$$

$$p = -6$$

FINAL ANSWER

$p = -6, \ q = -2$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2022 | Q#: 22 | Marks: 7

TOPIC: **Coordinate Geometry**

Source: Jun 2022 P2H

22 *ABCD* is a kite, with diagonals *AC* and *BD*, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates (−3, 4)

The diagonals of the kite intersect at the point *M* with coordinates (0, 2)

Given that $AB = AD = 6.5$ cm and the x coordinate of *B* is positive,

find the coordinates of the points *B* and *D*.

(.......................... , ...)

(.......................... , ...)

(Total for Question 22 is 7 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jun 2022 P2H | Question 22
Coordinate Geometry

Topic check. Correct.

Method

The midpoint of BD is $M = (0, 2)$.
Since $A = (-3, 4)$, the gradient of AM is

$$\frac{2-4}{0-(-3)} = -\frac{2}{3}$$

For the kite, diagonal AC is perpendicular to BD, so the gradient of BD is

$$\frac{3}{2}$$

Use direction vector $(2, 3)$ along BD.
Let

$$B = (2t, 2+3t)$$

and

$$D = (-2t, 2-3t)$$

Since $AB = 6.5$,

$$(2t-(-3))^2 + (2+3t-4)^2 = 6.5^2$$

$$(2t+3)^2 + (3t-2)^2 = 42.25$$

$$4t^2 + 12t + 9 + 9t^2 - 12t + 4 = 42.25$$

$$13t^2 + 13 = 42.25$$

$$13t^2 = 29.25$$

$$t^2 = 2.25$$

$$t = 1.5$$

The x-coordinate of B is positive, so use $t = 1.5$.

$$B = (3, 6.5)$$

$$D = (-3, -2.5)$$

FINAL ANSWER
$B = (3, 6.5), \ D = (-3, -2.5)$

CLASSIFIED PROBLEM Paper: P2HR | Session: Jun 2022 | Q#: 20 | Marks: 4

TOPIC: **Coordinate Geometry**

Source: Jun 2022 P2HR

20 The centre O of a circle has coordinates (4, 7)

The point A, on the circle, has coordinates (6, 11) and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

....

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 20
Coordinate Geometry

Topic check. Correct.

Method

The centre of the circle is

$$O = (4, 7)$$

The point A is

$$A = (6, 11)$$

Since AOP is a diameter, O is the midpoint of AP.
Let $P = (x, y)$.

$$\left(\frac{6+x}{2}, \frac{11+y}{2}\right) = (4, 7)$$

$$6 + x = 8, \qquad 11 + y = 14$$

$$P = (2, 3)$$

The gradient of OP is

$$\frac{3-7}{2-4} = 2$$

The tangent is perpendicular to the radius, so its gradient is

$$-\frac{1}{2}$$

Using point $P = (2, 3)$,

$$y - 3 = -\frac{1}{2}(x - 2)$$

$$y = -\frac{1}{2}x + 4$$

FINAL ANSWER
$y = -\frac{1}{2}x + 4$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2023 | Q#: 24 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jun 2023 P2H

24 $ABCD$ is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(.................... , ...)

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 24

Coordinate Geometry

Topic check. Correct.

Method

In the kite, $AB = AD$ and $CB = CD$, so AC is the line of symmetry.
Therefore D is the reflection of B in the line AC.

$$A = (-2, 10), \qquad B = \left(-\frac{27}{5}, 4\right), \qquad C = (4, -5)$$

A direction vector for AC is

$$\begin{pmatrix} 4-(-2) \\ -5-10 \end{pmatrix} = \begin{pmatrix} 6 \\ -15 \end{pmatrix}$$

Use the simpler direction vector

$$\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

Let H be the foot of the perpendicular from B to AC.

$$\vec{AB} = \begin{pmatrix} -\frac{17}{5} \\ -6 \end{pmatrix}$$

The fraction along AC to H is

$$\frac{\vec{AB} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix}}{2^2 + (-5)^2} = \frac{-\frac{34}{5} + 30}{29}$$

$$= \frac{\frac{116}{5}}{29} = \frac{4}{5}$$

So

$$H = A + \frac{4}{5}\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

$$H = \left(-2 + \frac{8}{5}, 10 - 4\right)$$

$$H = \left(-\frac{2}{5}, 6\right)$$

Since H is the midpoint of BD,

$$D = 2H - B$$

$$D = \left(-\frac{4}{5}, 12\right) - \left(-\frac{27}{5}, 4\right)$$

$$D = \left(\frac{23}{5}, 8\right)$$

FINAL ANSWER

$D = \left(\frac{23}{5}, 8\right)$

CLASSIFIED PROBLEM Paper: P2HR | Session: Jun 2023 | Q#: 21 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jun 2023 P2HR

21 Work out the coordinates of the points of intersection of

$$y - 2x = 1 \quad \text{and} \quad y^2 + xy = 7$$

Show clear algebraic working.

(........................ , ...)

(........................ , ...)

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 21
Coordinate Geometry

Topic check. Correct.

Method

The line is

$$y - 2x = 1$$

So

$$y = 2x + 1$$

Substitute into

$$y^2 + xy = 7$$

$$(2x+1)^2 + x(2x+1) = 7$$

$$4x^2 + 4x + 1 + 2x^2 + x = 7$$

$$6x^2 + 5x - 6 = 0$$

Factorise:

$$(3x-2)(2x+3) = 0$$

So

$$x = \frac{2}{3} \quad \text{or} \quad x = -\frac{3}{2}$$

Using $y = 2x + 1$:

$$x = \frac{2}{3} \Rightarrow y = \frac{7}{3}$$

$$x = -\frac{3}{2} \Rightarrow y = -2$$

FINAL ANSWER
$\left(\frac{2}{3}, \frac{7}{3}\right)$ and $\left(-\frac{3}{2}, -2\right)$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2023 | Q#: 25 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jun 2023 P2HR

25 The straight line with equation $y - 2x = 7$ is the perpendicular bisector of the line AB where A is the point with coordinates $(j, 7)$ and B is the point with coordinates $(6, k)$

Find the coordinates of the midpoint of the line AB
Show clear algebraic working.

(.......................... , ...)

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 25
Coordinate Geometry

Topic check. Moved to Coordinate Geometry.

Method

The perpendicular bisector is

$$y - 2x = 7$$

so

$$y = 2x + 7$$

The gradient of the perpendicular bisector is 2, so the gradient of AB is

$$-\frac{1}{2}$$

Since $A = (j, 7)$ and $B = (6, k)$,

$$\frac{k-7}{6-j} = -\frac{1}{2}$$

$$2(k-7) = -(6-j)$$

$$j = 2k - 8$$

The midpoint of AB is

$$\left(\frac{j+6}{2}, \frac{7+k}{2}\right)$$

Substitute $j = 2k - 8$:

$$\left(k-1, \frac{k+7}{2}\right)$$

The midpoint lies on $y = 2x + 7$, so

$$\frac{k+7}{2} = 2(k-1) + 7$$

$$k + 7 = 4k + 10$$

$$k = -1$$

Then

$$j = 2(-1) - 8 = -10$$

Therefore the midpoint is

$$\left(\frac{-10+6}{2}, \frac{7+(-1)}{2}\right) = (-2, 3)$$

FINAL ANSWER

$\boxed{(-2, 3)}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2024 | Q#: 22 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jun 2024 P2H

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.......................... , ...) (.......................... , ...)

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 22
Coordinate Geometry

Topic check. Correct.

Method

The line is

$$x + y = 5$$

So

$$y = 5 - x$$

Substitute into the curve:

$$2x^2 + 3y^2 = 210$$

$$2x^2 + 3(5 - x)^2 = 210$$

$$2x^2 + 3(x^2 - 10x + 25) = 210$$

$$5x^2 - 30x + 75 = 210$$

$$5x^2 - 30x - 135 = 0$$

Divide by 5:

$$x^2 - 6x - 27 = 0$$

Factorise:

$$(x - 9)(x + 3) = 0$$

So

$$x = 9 \quad \text{or} \quad x = -3$$

Using $y = 5 - x$:

$$x = 9 \Rightarrow y = -4$$

$$x = -3 \Rightarrow y = 8$$

FINAL ANSWER
$(9, -4)$ and $(-3, 8)$

CLASSIFIED PROBLEM Paper: P2HR | Session: Jun 2024 | Q#: 21 | Marks: 4

TOPIC: **Coordinate Geometry**

Source: Jun 2024 P2HR

21 *ABCD* is a square.

The point A has coordinates $(-5, 2)$
The point B has coordinates $(3, 5)$

Find an equation of the line that passes through B and C
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

....

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 21
Coordinate Geometry

Topic check. Correct.

Method

$$A = (-5, 2), \qquad B = (3, 5)$$

The gradient of AB is

$$\frac{5-2}{3-(-5)} = \frac{3}{8}$$

Since $ABCD$ is a square, BC is perpendicular to AB.
So the gradient of BC is

$$-\frac{8}{3}$$

The line through $B = (3, 5)$ is

$$y - 5 = -\frac{8}{3}(x-3)$$

Multiply by 3:

$$3y - 15 = -8x + 24$$

$$8x + 3y - 39 = 0$$

FINAL ANSWER
$8x + 3y - 39 = 0$

CLASSIFIED PROBLEM

Paper: P1H | Session: Jun 2025 | Q#: 25 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jun 2025 P1H

25 *PQRS* is a square.

PR is a diagonal of the square.

P is the point with coordinates (4, 7)
R is the point with coordinates (8, –5)

Find an equation of the straight line that passes through the points Q and S
Give your answer in the form $ay = bx + c$ where a, b and c are integers.

....

(Total for Question 25 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 25
Coordinate Geometry

Topic check. Correct.

Method

$$P = (4,7), \qquad R = (8,-5)$$

The gradient of diagonal PR is

$$\frac{-5-7}{8-4} = -3$$

The other diagonal QS is perpendicular to PR, so its gradient is

$$\frac{1}{3}$$

The diagonals of a square bisect each other.
The midpoint of PR is

$$\left(\frac{4+8}{2}, \frac{7+(-5)}{2}\right) = (6,1)$$

So QS passes through $(6,1)$ with gradient $\frac{1}{3}$.

$$y - 1 = \frac{1}{3}(x-6)$$

$$y = \frac{1}{3}x - 1$$

Multiply by 3:

$$3y = x - 3$$

FINAL ANSWER
$3y = x - 3$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2025 | Q#: 24 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jun 2025 P2HR

24 PQ is a straight line drawn on a square grid, with a scale of 1 cm for 1 unit on each axis.

P has coordinates $(-5,\ a)$ and Q has coordinates $(7,\ 3a)$ where $a > 0$

The length of PQ is $4\sqrt{10}$ cm

Find an equation of the perpendicular bisector of PQ
Give your answer in the form $y = mx + c$ where m and c are integers.

....

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 24
Coordinate Geometry

Topic check. Moved to Coordinate Geometry.

Method

$$P = (-5, a), \qquad Q = (7, 3a)$$

The length of PQ is $4\sqrt{10}$, so

$$\sqrt{(7-(-5))^2 + (3a-a)^2} = 4\sqrt{10}$$

$$\sqrt{12^2 + (2a)^2} = 4\sqrt{10}$$

Square both sides:

$$144 + 4a^2 = 160$$

$$4a^2 = 16$$

$$a^2 = 4$$

Since $a > 0$,

$$a = 2$$

So

$$P = (-5, 2), \qquad Q = (7, 6)$$

The midpoint of PQ is

$$\left(\frac{-5+7}{2}, \frac{2+6}{2}\right) = (1, 4)$$

The gradient of PQ is

$$\frac{6-2}{7-(-5)} = \frac{4}{12} = \frac{1}{3}$$

So the gradient of the perpendicular bisector is

$$-3$$

Using point $(1, 4)$:

$$y - 4 = -3(x - 1)$$

$$y = -3x + 7$$

FINAL ANSWER

$\boxed{y = -3x + 7}$

CLASSIFIED PROBLEM

Paper: P1H **Session: MayNov 2020** **Q#: 22** **Marks: 4**

TOPIC: **Coordinate Geometry**

Source: May-Nov 2020 P1H

22 *ABCD* is a rhombus.

The diagonals, *AC* and *BD*, intersect at the point *M*.
The coordinates of *M* are $(6, -11)$

The points *A* and *C* both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through *B* and *D* intersects the *y*-axis.

(.......................... , ...)

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 22
Coordinate Geometry

Topic check. Correct.

Method

In a rhombus, the diagonals are perpendicular and bisect each other.
The diagonal AC lies on

$$2y + 7x = 20$$

$$2y = -7x + 20$$

$$y = -\frac{7}{2}x + 10$$

So the gradient of AC is

$$-\frac{7}{2}$$

Therefore the gradient of BD is

$$\frac{2}{7}$$

The line BD passes through

$$M = (6, -11)$$

So

$$y + 11 = \frac{2}{7}(x - 6)$$

To find where this line intersects the y-axis, put $x = 0$:

$$y + 11 = \frac{2}{7}(0 - 6)$$

$$y + 11 = -\frac{12}{7}$$

$$y = -11 - \frac{12}{7}$$

$$y = -\frac{89}{7}$$

FINAL ANSWER

$$\left(0, -\frac{89}{7}\right)$$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2021 | Q#: 20 | Marks: 7

TOPIC: **Coordinate Geometry**

Source: Nov 2021 P1H

20 The straight line **L** passes through point A (– 6, 2) and point B (5, 3)
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B.
The line **M** intersects the line $x = -1$ at point C.

Calculate the area of triangle ABC.

....

(Total for Question 20 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2021 P1H | Question 20

Coordinate Geometry

Topic check. Correct.

Method

$$A = (-6, 2), \qquad B = (5, 3)$$

The midpoint of AB is

$$\left(\frac{-6+5}{2}, \frac{2+3}{2}\right) = \left(-\frac{1}{2}, \frac{5}{2}\right)$$

The gradient of AB is

$$\frac{3-2}{5-(-6)} = \frac{1}{11}$$

So the perpendicular gradient is

$$-11$$

Line M passes through $\left(-\frac{1}{2}, \frac{5}{2}\right)$, so

$$y - \frac{5}{2} = -11\left(x + \frac{1}{2}\right)$$

At point C, $x = -1$:

$$y - \frac{5}{2} = -11\left(-\frac{1}{2}\right)$$

$$y - \frac{5}{2} = \frac{11}{2}$$

$$y = 8$$

So

$$C = (-1, 8)$$

Now use vectors from A:

$$\vec{AB} = (11, 1), \qquad \vec{AC} = (5, 6)$$

Area of triangle ABC is

$$\frac{1}{2}|11(6) - 1(5)|$$

$$= \frac{1}{2}(61)$$

$$= 30.5$$

FINAL ANSWER

30.5

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2023 | Q#: 22 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Nov 2023 P1H

22 [In this question 1 cm = 1 unit on the x-axis and 1 cm = 1 unit on the y-axis]

P is a point on a circle with centre (0, 0)
The coordinates of P are (8, −10)

The line **L** is the tangent to the circle at the point P
L crosses the x-axis at the point Q and crosses the y-axis at the point R

Work out the length of RQ
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P1H | Question 22
Coordinate Geometry

Topic check. Correct.

Method

The centre of the circle is $(0, 0)$, and

$$P = (8, -10)$$

The gradient of OP is

$$\frac{-10}{8} = -\frac{5}{4}$$

So the gradient of the tangent is

$$\frac{4}{5}$$

The tangent through P is

$$y + 10 = \frac{4}{5}(x - 8)$$

$$y = \frac{4}{5}x - \frac{82}{5}$$

At the x-axis, $y = 0$:

$$0 = \frac{4}{5}x - \frac{82}{5}$$

$$x = \frac{41}{2}$$

So

$$Q = \left(\frac{41}{2}, 0\right)$$

At the y-axis, $x = 0$:

$$R = \left(0, -\frac{82}{5}\right)$$

Therefore

$$RQ = \sqrt{\left(\frac{41}{2}\right)^2 + \left(\frac{82}{5}\right)^2}$$

$$RQ = 26.252...$$

FINAL ANSWER
26.3 cm

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2023 | Q#: 21 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Nov 2023 P2H

21 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | Question 21
Coordinate Geometry

Topic check. Correct.

Method

The line is

$$x + 2y = 5$$

So

$$x = 5 - 2y$$

Substitute into

$$x^2 + 3y^2 = 13$$

$$(5 - 2y)^2 + 3y^2 = 13$$

$$25 - 20y + 4y^2 + 3y^2 = 13$$

$$7y^2 - 20y + 12 = 0$$

Factorise:

$$(7y - 6)(y - 2) = 0$$

So

$$y = \frac{6}{7} \quad \text{or} \quad y = 2$$

Using $x = 5 - 2y$:

$$y = \frac{6}{7} \Rightarrow x = \frac{23}{7}$$

$$y = 2 \Rightarrow x = 1$$

FINAL ANSWER
$\left(\frac{23}{7}, \frac{6}{7}\right)$ and $(1, 2)$

CLASSIFIED PROBLEM Paper: P1H | Session: Nov 2025 | Q#: 20 | Marks: 3

TOPIC: **Coordinate Geometry**

Source: Nov 2025 P1H

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**
Give your answer in the form $y = mx + c$

....

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 20

Coordinate Geometry

Topic check. Correct.

Method

Line L has equation

$$y = 4x + 7$$

So its gradient is

$$4$$

Line M is perpendicular to L, so its gradient is

$$-\frac{1}{4}$$

Line M passes through $(8, 1)$.

$$y - 1 = -\frac{1}{4}(x - 8)$$

$$y - 1 = -\frac{1}{4}x + 2$$

$$y = -\frac{1}{4}x + 3$$

FINAL ANSWER

$y = -\frac{1}{4}x + 3$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2025 | Q#: 28 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Nov 2025 P1H

28 The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are (0.2, 2.6)

Work out the coordinates of B
Show your working clearly.

(.................... , ...)

(Total for Question 28 is 6 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Nov 2025 P1H | Question 28
Coordinate Geometry

Topic check. Correct.

Method
The line is

$$y = 3x + e$$

Point $A = (0.2, 2.6)$ lies on the line.

$$2.6 = 3(0.2) + e$$

$$2.6 = 0.6 + e$$

$$e = 2$$

So

$$y = 3x + 2$$

The curve is

$$x^2 + y^2 = d - 11x$$

Use point $A = (0.2, 2.6)$:

$$0.2^2 + 2.6^2 = d - 11(0.2)$$

$$0.04 + 6.76 = d - 2.2$$

$$d = 9$$

Now substitute $y = 3x + 2$ into the curve:

$$x^2 + (3x + 2)^2 = 9 - 11x$$

$$x^2 + 9x^2 + 12x + 4 = 9 - 11x$$

$$10x^2 + 23x - 5 = 0$$

One root is $0.2 = \frac{1}{5}$.
The product of the roots is

$$\frac{-5}{10} = -\frac{1}{2}$$

So the other root is

$$\frac{-\frac{1}{2}}{\frac{1}{5}} = -\frac{5}{2}$$

Then

$$y = 3\left(-\frac{5}{2}\right) + 2$$

$$y = -\frac{11}{2}$$

FINAL ANSWER
$B = \left(-\frac{5}{2}, -\frac{11}{2}\right)$

Topic 10

Graphs of Functions

CLASSIFIED PROBLEM

Paper: P1H | Session: Jan 2020 | Q#: 20 | Marks: 4

TOPIC: **Graphs of Functions**

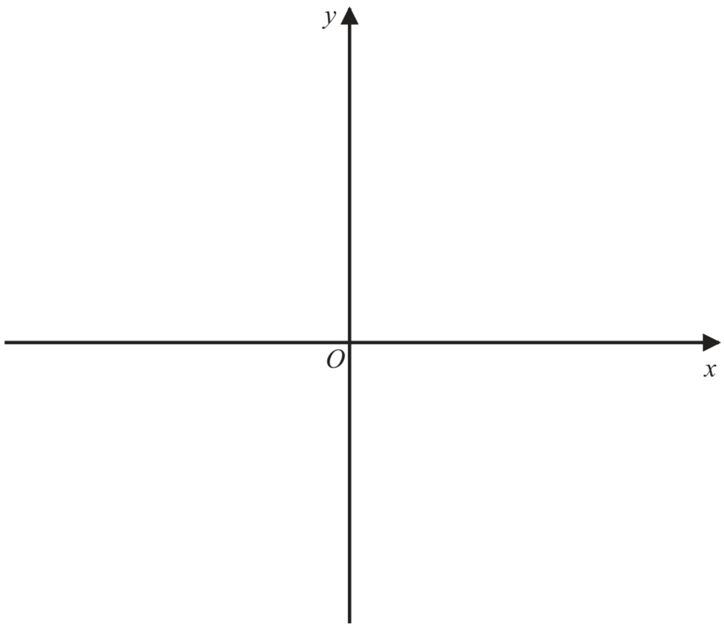
Source: Jan 2020 P1H

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of a,

(i) the coordinates of any points of intersection of **C** with the coordinate axes,

(ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1H | Question 20

Graphs of Functions

Topic check. Correct.

Method

$$y = 4(x-1)^2 - a$$

The turning point is read directly from the completed square form:

$$(1, -a)$$

For the y-intercept, put $x = 0$:

$$y = 4(0-1)^2 - a$$

$$y = 4 - a$$

So the y-intercept is

$$(0, 4-a)$$

For the x-intercepts, put $y = 0$:

$$4(x-1)^2 - a = 0$$

$$4(x-1)^2 = a$$

$$(x-1)^2 = \frac{a}{4}$$

$$x - 1 = \pm\frac{\sqrt{a}}{2}$$

$$x = 1 \pm \frac{\sqrt{a}}{2}$$

So the x-intercepts are

$$\left(1 - \frac{\sqrt{a}}{2}, 0\right) \quad \text{and} \quad \left(1 + \frac{\sqrt{a}}{2}, 0\right)$$

Since $a > 4$, the turning point is below the x-axis and the curve crosses the x-axis twice.

FINAL ANSWER

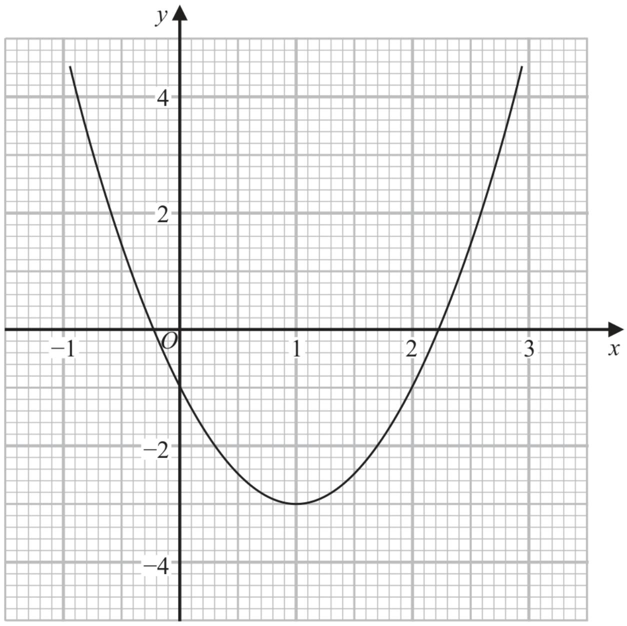
turning point $(1, -a)$, y-intercept $(0, 4-a)$, x-intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

CLASSIFIED PROBLEM Paper: P2H | Session: Jan 2022 | Q#: 21 | Marks: 5

TOPIC: **Graphs of Functions**

Source: Jan 2022 P2H

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....

(2)

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....

(3)

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 21

Graphs of Functions

Topic check. Moved from Linear Graphs to Graphs of Functions.

Method

For part (a), solve

$$2x^2 - 4x - 1 = 0$$

Using the graph gives approximately

$$x = -0.2, \quad x = 2.2$$

For part (b), we need to solve

$$x^2 - x - 1 = 0$$

Multiply by 2:

$$2x^2 - 2x - 2 = 0$$

The given graph is

$$y = 2x^2 - 4x - 1$$

Now

$$2x^2 - 2x - 2 = (2x^2 - 4x - 1) + 2x - 1$$

So draw the straight line

$$y = 1 - 2x$$

The intersections of this line with the curve give approximately

$$x = -0.6, \quad x = 1.6$$

FINAL ANSWER

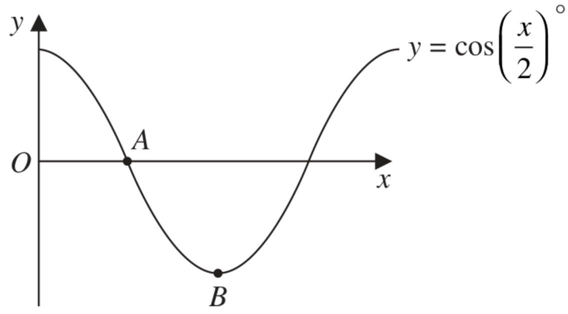
(a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

CLASSIFIED PROBLEM | Paper: P2H | Session: Jun 2022 | Q#: 23 | Marks: 2

TOPIC: **Graphs of Functions**

Source: Jun 2022 P2H

23 The diagram shows a sketch of the graph of $y = \cos\left(\dfrac{x}{2}\right)^{\circ}$



(i) Find the coordinates of the point A

(.......................... , ...)

(1)

(ii) Find the coordinates of the point B

(.......................... , ...)

(1)

(Total for Question 23 is 2 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jun 2022 P2H | Question 23
Graphs of Functions

Topic check. Moved from Coordinate Geometry to Graphs of Functions.

Method

The graph is

$$y = \cos\left(\frac{x}{2}\right)^{\circ}$$

Point A is the first positive x-intercept, so $y = 0$.

$$\cos\left(\frac{x}{2}\right) = 0$$

$$\frac{x}{2} = 90^{\circ}$$

$$x = 180$$

So

$$A = (180, 0)$$

Point B is the minimum point.
The minimum value of cosine is -1, which occurs when

$$\frac{x}{2} = 180^{\circ}$$

$$x = 360$$

So

$$B = (360, -1)$$

FINAL ANSWER
$A = (180, 0),\ B = (360, -1)$

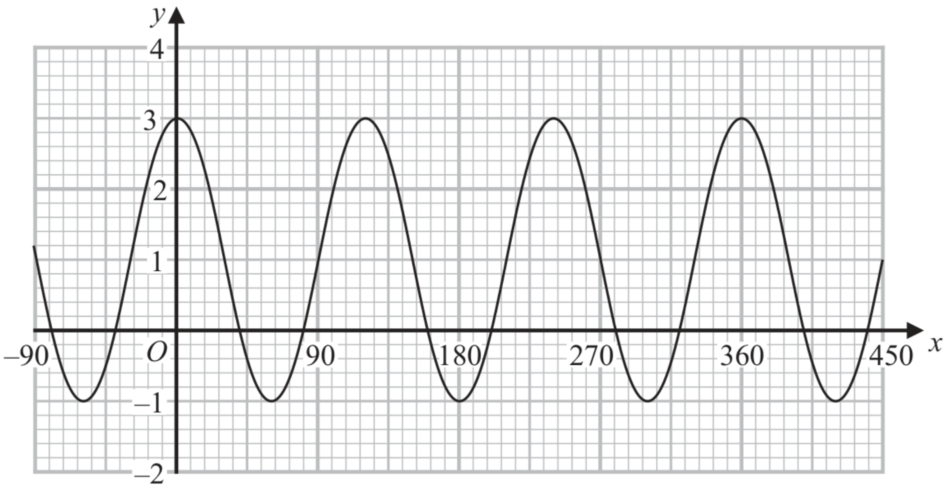
CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2023 | Q#: 26 | Marks: 3

TOPIC: **Graphs of Functions**

Source: Jun 2023 P2HR

26 Here is a sketch of the curve with equation $y = a\cos bx° + c$ where $-90 \leqslant x \leqslant 450$



Find the value of a, the value of b and the value of c

a = ...

b = ...

c = ...

(Total for Question 26 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 26

Graphs of Functions

Topic check. Corrected from Sine/Cosine Rule to Graphs of Functions.

Method

The curve is

$$y = a\cos bx^\circ + c$$

From the graph, the maximum value is

$$3$$

and the minimum value is

$$-1$$

The amplitude is half the difference:

$$a = \frac{3-(-1)}{2} = 2$$

The midline is the average:

$$c = \frac{3+(-1)}{2} = 1$$

The distance between consecutive maxima is 120°, so the period is

$$120^\circ$$

For $y = a\cos bx^\circ + c$, the period is

$$\frac{360^\circ}{b}$$

So

$$\frac{360}{b} = 120$$

$$b = 3$$

FINAL ANSWER

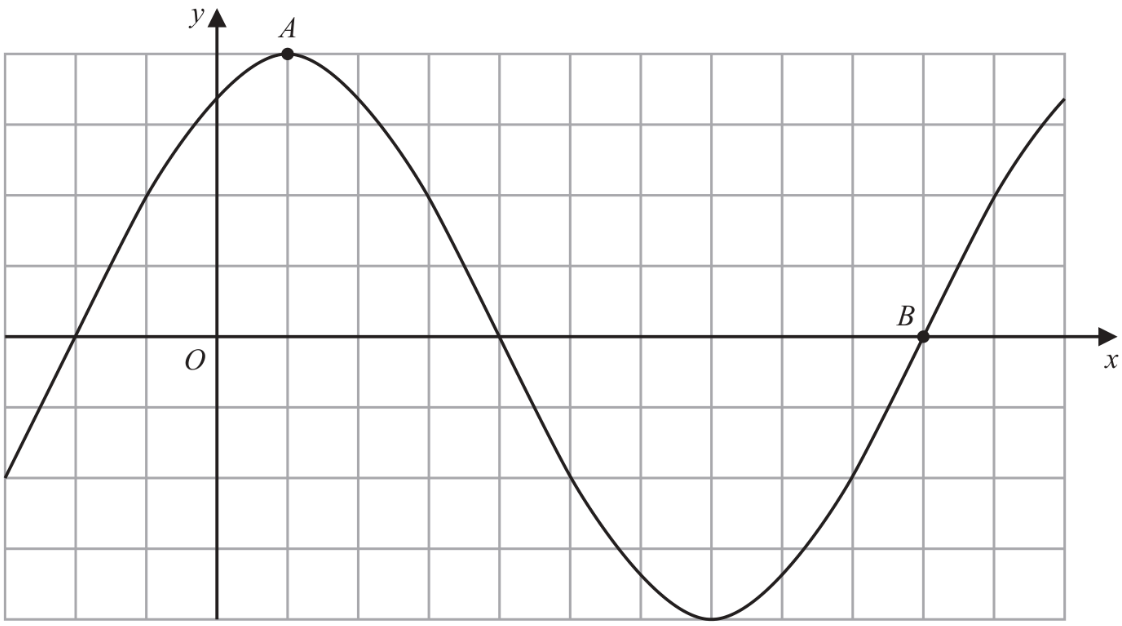
$\boxed{a = 2,\ b = 3,\ c = 1}$

CLASSIFIED PROBLEM Paper: P2H Session: Jun 2024 Q#: 25 Marks: 2

TOPIC: **Graphs of Functions**

Source: Jun 2024 P2H

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.......................... , ...)
(1)

(ii) Find the coordinates of the point B

(.......................... , ...)
(1)

(Total for Question 25 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 25
Graphs of Functions

Topic check. Moved from Coordinate Geometry to Graphs of Functions.

Method

$$y = 2\sin(x + 60)^\circ$$

Point A is the maximum point.
The maximum value is

$$y = 2$$

This happens when

$$x + 60 = 90$$

$$x = 30$$

So

$$A = (30, 2)$$

Point B is the positive x-intercept after the minimum.

$$2\sin(x + 60)^\circ = 0$$

$$\sin(x + 60)^\circ = 0$$

For this point,

$$x + 60 = 360$$

$$x = 300$$

So

$$B = (300, 0)$$

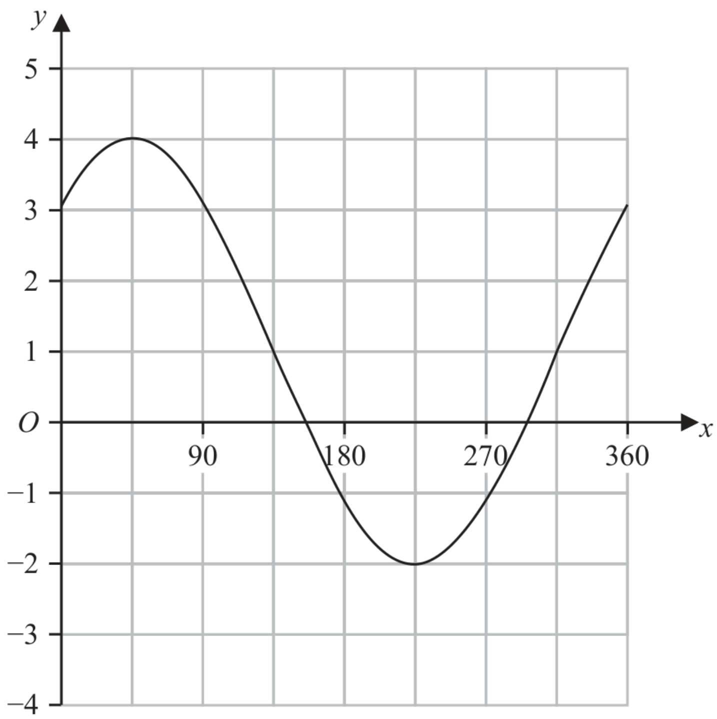
FINAL ANSWER
$A = (30, 2),\ B = (300, 0)$

CLASSIFIED PROBLEM Paper: P2HR Session: Jun 2025 Q#: 25 Marks: 3

TOPIC: **Graphs of Functions**

Source: Jun 2025 P2HR

25 The graph of $y = a\sin(x + b)^\circ + c$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Find a suitable value for a, for b and for c

a = ...

b = ...

c = ...

(Total for Question 25 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 25

Graphs of Functions

Topic check. Correct.

Method

From the graph,

$$\text{maximum} = 4$$

$$\text{minimum} = -2$$

The amplitude is

$$a = \frac{4 - (-2)}{2} = 3$$

The centre line is

$$c = \frac{4 + (-2)}{2} = 1$$

The maximum is about 45°. For a sine graph, a maximum happens when the angle inside the sine is 90°, so

$$45 + b = 90$$

$$b = 45$$

FINAL ANSWER

$a = 3,\ b = 45,\ c = 1$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2024 | Q#: 23 | Marks: 5

TOPIC: **Graphs of Functions**

Source: Nov 2024 P1H

23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)
The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k
Show clear algebraic working.

k = ...

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | Question 23

Graphs of Functions

Topic check. Moved from Coordinate Geometry to Graphs of Functions.

Method

The curve is

$$y = x^2 - 8x - 9$$

The line is

$$y = k$$

At the intersection points,

$$x^2 - 8x - 9 = k$$

$$x^2 - 8x - (9 + k) = 0$$

The two roots are p and q.
For this quadratic,

$$p + q = 8$$

We are also given

$$p - q = 14$$

Solve the two equations:

$$2p = 22$$

$$p = 11$$

$$q = -3$$

Use one root in $y = x^2 - 8x - 9$:

$$k = 11^2 - 8(11) - 9$$

$$k = 121 - 88 - 9$$

$$k = 24$$

FINAL ANSWER

$k = 24$

Topic 11

Standard & Compound Units

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2022 | Q#: 21 | Marks: 6

TOPIC: **Standard & Compound Units**

Source: Jan 2022 P2HR

21 The diagram shows a solid made from a cylinder and a hemisphere.
The cylinder and the hemisphere are both made from the same metal.

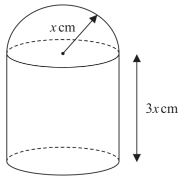


Diagram **NOT** accurately drawn

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The radius of the cylinder and the radius of the hemisphere are both x cm.
The height of the cylinder is $3x$ cm.

The total surface area of the solid is 81π cm^2
The mass of the solid is 840 grams.

The following table gives the density of each of four metals.

Metal	Density (g/cm^3)
Aluminium	2.7
Nickel	8.9
Gold	19.3
Silver	10.5

The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.
Show your working clearly.

....

(Total for Question 21 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2HR | Question 21
Standard & Compound Units

Topic check. Correct.

Method

The outside surface area is:
curved surface area of cylinder:

$$2\pi x(3x) = 6\pi x^2$$

bottom of cylinder:

$$\pi x^2$$

curved surface area of hemisphere:

$$2\pi x^2$$

So

$$6\pi x^2 + \pi x^2 + 2\pi x^2 = 81\pi$$

$$9\pi x^2 = 81\pi$$

$$x^2 = 9$$

$$x = 3$$

Volume of the cylinder:

$$\pi x^2(3x) = 3\pi x^3$$

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

Total volume:

$$3\pi(3)^3 + \frac{2}{3}\pi(3)^3$$

$$= 81\pi + 18\pi = 99\pi$$

Density:

$$\frac{840}{99\pi} = 2.70\ldots$$

This matches aluminium, whose density is 2.7 g/cm^3.

FINAL ANSWER
Aluminium

CLASSIFIED PROBLEM Paper: P2H | Session: Nov 2025 | Q#: 26 | Marks: 6

TOPIC: **Standard & Compound Units**

Source: Nov 2025 P2H

26 A solid cone is joined to a solid hemisphere to make the solid shown below.

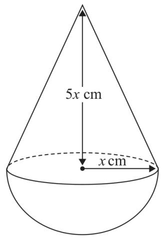


Diagram **NOT** accurately drawn

The cone is made from copper.
The density of copper is $9\,g/cm^3$

The hemisphere is made from a different metal.

The total mass of the solid is 4752π grams
The total volume of the solid is $504\pi\,cm^3$

Work out the density of the hemisphere.
Show your working clearly.

.... g/cm^3

(Total for Question 26 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 26

Standard & Compound Units

Topic check. Correct.

Method

Volume of cone:

$$\frac{1}{3}\pi x^2(5x) = \frac{5}{3}\pi x^3$$

Volume of hemisphere:

$$\frac{2}{3}\pi x^3$$

Total volume:

$$\frac{5}{3}\pi x^3 + \frac{2}{3}\pi x^3 = 504\pi$$

$$\frac{7}{3}\pi x^3 = 504\pi$$

$$x^3 = 216$$

$$x = 6$$

Cone volume:

$$\frac{5}{3}\pi(6)^3 = 360\pi$$

Mass of copper cone:

$$360\pi \times 9 = 3240\pi$$

Mass of hemisphere:

$$4752\pi - 3240\pi = 1512\pi$$

Hemisphere volume:

$$\frac{2}{3}\pi(6)^3 = 144\pi$$

Density of the hemisphere:

$$\frac{1512\pi}{144\pi} = 10.5$$

FINAL ANSWER

$10.5\ \text{g/cm}^3$

Topic 12

Area & Perimeter

CLASSIFIED PROBLEM Paper: 4WM1H Session: Nov2025 Q#: 21 Marks: 4

Topic: **Area & Perimeter**

Source: Nov 2025 4WM1H

21 The diagram shows an equilateral triangle and a circle.

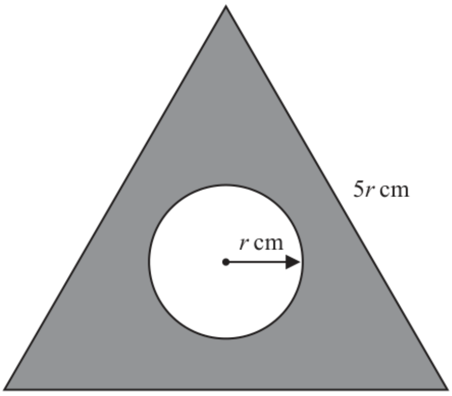


Diagram **NOT** accurately drawn

The equilateral triangle has sides of length $5r$ cm
The circle has radius r cm

The area of the shaded region is 610π cm^2
Work out the value of r

Give your answer correct to 3 significant figures.

r = ...

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 21

Area & Perimeter

Solution

Triangle area:

$$\frac{\sqrt{3}}{4}(5r)^2 = \frac{25\sqrt{3}}{4}r^2$$

Circle area:

$$\pi r^2$$

So

$$\left(\frac{25\sqrt{3}}{4} - \pi\right) r^2 = 610\pi$$

$$r \approx 15.7926$$

FINAL ANSWER

$r = 15.8$ to 3 significant figures.

Topic 13

Circles, Arcs & Sectors

CLASSIFIED PROBLEM

Paper: 4WM1HR | Session: May2025 | Q#: 22 | Marks: 6

TOPIC: Circles, Arcs & Sectors

Source: May 2025 4WM1HR

22

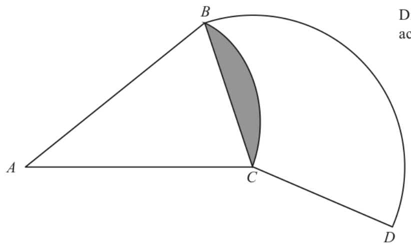


Diagram **NOT** accurately drawn

BAC is a sector of a circle, centre *A*

BCD is a sector of a circle, centre *C*

Angle $BAC = 40°$
Angle $BCD = 130°$
Area of shaded segment = $28\,\text{cm}^2$

Find the length of the arc *BD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 22
Circles, Arcs & Sectors

Topic check. Correct.

Method

Let the radius of sector BAC be R.
The shaded segment is the segment cut off by chord BC in the sector with centre A.
Its area is

$$\text{sector area} - \text{triangle area}$$

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40^\circ$$

$$28 = R^2\left(\frac{\pi}{9} - \frac{1}{2}\sin 40^\circ\right)$$

$$R = 31.8096\ldots$$

Now find the chord BC:

$$BC = 2R\sin 20^\circ$$

$$BC = 2(31.8096\ldots)\sin 20^\circ = 21.7591\ldots$$

In sector BCD, the centre is C, so the radius is

$$CB = CD = 21.7591\ldots$$

The arc BD has angle 130°, so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591\ldots)$$

$$= 49.3697\ldots$$

FINAL ANSWER

$\boxed{49.4 \text{ cm}}$

Topic 14

Area & Perimeter

CLASSIFIED PROBLEM Paper: P2H | Session: Jan 2021 | Q#: 20 | Marks: 4

TOPIC: **Area & Perimeter**

Source: Jan 2021 P2H

20

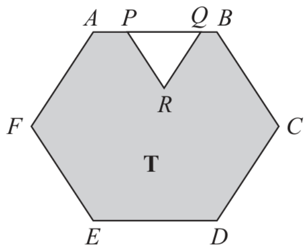


Diagram **NOT** accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3}$ cm²

work out the length of PQ.

.... cm

(Total for Question 20 is 4 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jan 2021 P2H | Question 20
Area & Perimeter

Topic check. Moved from Angles in Polygons & Parallel Lines to Area & Perimeter.

Method

Let

$$PQ = p$$

Given

$$AB = 1.5PQ$$

so the side length of the regular hexagon is

$$1.5p = \frac{3}{2}p$$

Area of a regular hexagon with side length s is

$$\frac{3\sqrt{3}}{2}s^2$$

So the area of the hexagon is

$$\frac{3\sqrt{3}}{2}\left(\frac{3}{2}p\right)^2$$

$$= \frac{27\sqrt{3}}{8}p^2$$

The removed triangle PQR is equilateral, so its area is

$$\frac{\sqrt{3}}{4}p^2$$

The shaded area is

$$\frac{27\sqrt{3}}{8}p^2 - \frac{\sqrt{3}}{4}p^2$$

$$= \frac{25\sqrt{3}}{8}p^2$$

This is $72\sqrt{3}$, so

$$\frac{25\sqrt{3}}{8}p^2 = 72\sqrt{3}$$

$$\frac{25}{8}p^2 = 72$$

$$p^2 = \frac{576}{25}$$

$$p = 4.8$$

FINAL ANSWER

$PQ = 4.8$ cm

CLASSIFIED PROBLEM

Paper: P1HR | Session: MayNov 2020 | Q#: 22 | Marks: 6

TOPIC: **Area & Perimeter**

Source: May-Nov 2020 P1HR

22 The diagram shows a regular octagon *ABCDEFGH*.

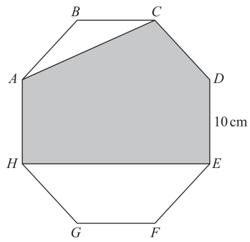


Diagram **NOT** accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region *ACDEH*.
Give your answer correct to the nearest cm^2

.... cm^2

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | Question 22

Area & Perimeter

Topic check. Correct.

Method

Each side of the regular octagon is 10 cm.
For a regular octagon with side length s,

$$\text{area} = 2(1+\sqrt{2})s^2$$

So the area of the whole octagon is

$$2(1+\sqrt{2})(10)^2 = 200 + 200\sqrt{2}$$

The unshaded triangle ABC has

$$AB = 10, \qquad BC = 10, \qquad \angle ABC = 135^\circ$$

So

$$\text{area of triangle } ABC = \frac{1}{2}(10)(10)\sin 135^\circ = 25\sqrt{2}$$

At the bottom, the unshaded trapezium $HEFG$ has parallel sides

$$HE = 10 + 10\sqrt{2}, \qquad GF = 10$$

and height

$$5\sqrt{2}$$

So

$$\text{area of trapezium } HEFG = \frac{1}{2}(10 + 10 + 10\sqrt{2})(5\sqrt{2})$$

$$= 50 + 50\sqrt{2}$$

The shaded area is

$$(200 + 200\sqrt{2}) - 25\sqrt{2} - (50 + 50\sqrt{2})$$

$$= 150 + 125\sqrt{2}$$

$$= 326.7766\ldots$$

FINAL ANSWER

$\boxed{327 \text{ cm}^2}$

Topic 15

Circles, Arcs & Sectors

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2021 | Q#: 22 | Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: Jan 2021 P2H

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

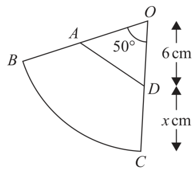


Diagram **NOT** accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50°$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x.
Give your answer correct to 3 significant figures.

$x =$ ....

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2H | Question 22

Circles, Arcs & Sectors

Topic check. Correct.

Method

The radius of sector OBC is

$$6 + x$$

The angle is $50°$, so the arc length of sector OBC is

$$\frac{50}{360} \times 2\pi(6 + x)$$

Therefore the perimeter of sector OBC is

$$2(6 + x) + \frac{50}{360} \times 2\pi(6 + x)$$

$$= (6 + x)\left(2 + \frac{5\pi}{18}\right)$$

In triangle OAD,

$$OA = OD = 6, \qquad \angle AOD = 50°$$

So

$$AD = 2(6)\sin 25° = 12 \sin 25°$$

The perimeter of triangle OAD is

$$12 + 12 \sin 25°$$

Given that

$$\text{perimeter of sector } OBC = 2 \times \text{perimeter of triangle } OAD$$

$$(6 + x)\left(2 + \frac{5\pi}{18}\right) = 2(12 + 12 \sin 25°)$$

$$x = \frac{2(12 + 12 \sin 25°)}{2 + \frac{5\pi}{18}} - 6$$

$$x = 5.8854\ldots$$

FINAL ANSWER

$\boxed{5.89}$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2021 | Q#: 20 | Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Source: Jan 2021 P2HR

20 A, B and C are points on a circle with centre O.

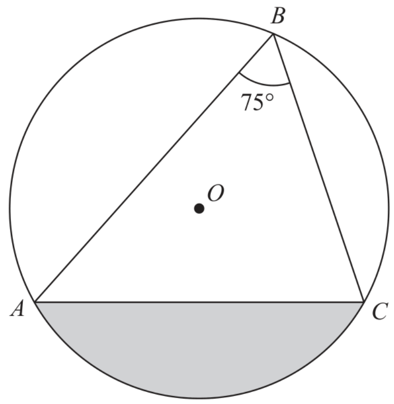


Diagram **NOT** accurately drawn

Angle $ABC = 75°$

The area of the shaded segment is $200\,\text{cm}^2$

Calculate the radius of the circle.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2HR | Question 20

Circles, Arcs & Sectors

Topic check. Moved to Circles, Arcs & Sectors.

Method

$$\angle ABC = 75^\circ$$

The angle at the centre is twice the angle at the circumference, so

$$\angle AOC = 2(75^\circ) = 150^\circ$$

The shaded segment is:

$$\text{sector } AOC - \text{triangle } AOC$$

Let the radius be r.
Sector area:

$$\frac{150}{360}\pi r^2$$

Triangle area:

$$\frac{1}{2}r^2 \sin 150^\circ$$

So

$$\frac{150}{360}\pi r^2 - \frac{1}{2}r^2 \sin 150^\circ = 200$$

$$r^2\left(\frac{5\pi}{12} - \frac{1}{4}\right) = 200$$

$$r^2 = \frac{200}{\frac{5\pi}{12} - \frac{1}{4}}$$

$$r = 13.7426\ldots$$

FINAL ANSWER

$\boxed{13.7 \text{ cm}}$

CLASSIFIED PROBLEM Paper: P2H Session: Jun 2022 Q#: 20 Marks: 4

TOPIC: Circles, Arcs & Sectors

Source: Jun 2022 P2H

20 The diagram shows four identical circles drawn inside a square.

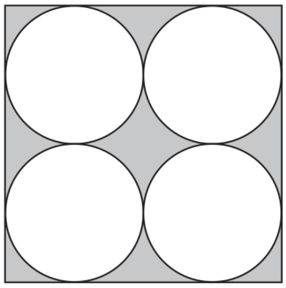


Diagram **NOT** accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 4 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jun 2022 P2H | Question 20
Circles, Arcs & Sectors

Topic check. Moved to Circles, Arcs & Sectors.

Method

Let the radius of each circle be r.
There are two circles across the width of the square, so the side length of the square is

$$4r$$

Area of the square:

$$(4r)^2 = 16r^2$$

Area of the four circles:

$$4\pi r^2$$

The shaded area is 40 cm^2, so

$$16r^2 - 4\pi r^2 = 40$$

$$r^2(16 - 4\pi) = 40$$

$$r = \sqrt{\frac{40}{16 - 4\pi}} = 3.4131\ldots$$

The perimeter of the square is

$$4(4r) = 16r$$

$$16r = 54.6101\ldots$$

FINAL ANSWER
$\boxed{54.6 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2024 | Q#: 21 | Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Source: Jun 2024 P2H

21 The diagram shows a square *ABCD* and a circle.

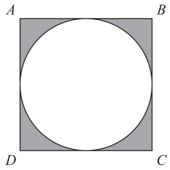


Diagram **NOT** accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is 80 cm^2

Work out the length of *AC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 21

Circles, Arcs & Sectors

Topic check. Moved to Circles, Arcs & Sectors.

Method

Let the side length of the square be s.

The circle is inscribed in the square, so its radius is

$$\frac{s}{2}$$

The shaded area is the area of the square minus the area of the circle:

$$s^2 - \pi\left(\frac{s}{2}\right)^2 = 80$$

$$s^2 - \frac{\pi s^2}{4} = 80$$

$$s^2\left(1 - \frac{\pi}{4}\right) = 80$$

$$s = \sqrt{\frac{80}{1 - \frac{\pi}{4}}} = 19.3076\ldots$$

The diagonal of the square is

$$AC = s\sqrt{2}$$

$$AC = 19.3076\ldots\sqrt{2} = 27.3051\ldots$$

FINAL ANSWER

$\boxed{27.3 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jun 2025 | Q#: 25 | Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: Jun 2025 P1HR

25

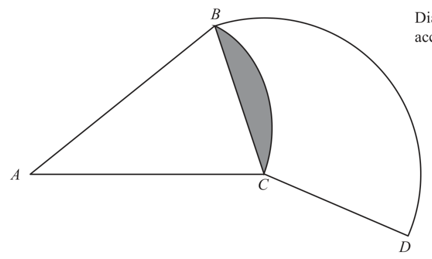


Diagram **NOT** accurately drawn

BAC is a sector of a circle, centre *A*

BCD is a sector of a circle, centre *C*

Angle $BAC = 40°$
Angle $BCD = 130°$
Area of shaded segment $= 28\,\text{cm}^2$

Find the length of the arc *BD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 25

Circles, Arcs & Sectors

Topic check. Correct.

Method

Let the radius of sector BAC be R.

The shaded segment is the segment cut off by chord BC in the sector with centre A.

Its area is

$$\text{sector area} - \text{triangle area}$$

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40^\circ$$

$$28 = R^2\left(\frac{\pi}{9} - \frac{1}{2}\sin 40^\circ\right)$$

$$R = 31.8096\ldots$$

Now find the chord BC:

$$BC = 2R\sin 20^\circ$$

$$BC = 2(31.8096\ldots)\sin 20^\circ = 21.7591\ldots$$

In sector BCD, the centre is C, so the radius is

$$CB = CD = 21.7591\ldots$$

The arc BD has angle 130°, so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591\ldots)$$

$$= 49.3697\ldots$$

FINAL ANSWER

$\boxed{49.4 \text{ cm}}$

CLASSIFIED PROBLEM

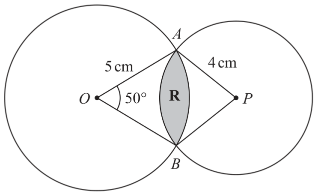
Paper: P1H | Session: May 2021 | Q#: 25 | Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: May 2021 P1H

25 The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

Diagram **NOT** accurately drawn



One of the circles has centre O and radius 5 cm.
The other circle has centre P and radius 4 cm.
Angle $AOB = 50°$

Calculate the area of region **R**.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 25
Circles, Arcs & Sectors

Topic check. Moved to Circles, Arcs & Sectors.

Method

For the circle with centre O:

$$\angle AOB = 50^\circ, \qquad OA = OB = 5$$

The half chord length is

$$\frac{AB}{2} = 5\sin 25^\circ$$

For the circle with centre P, let

$$\angle APB = \theta$$

The radius is 4, so

$$4\sin\left(\frac{\theta}{2}\right) = 5\sin 25^\circ$$

$$\sin\left(\frac{\theta}{2}\right) = \frac{5\sin 25^\circ}{4}$$

$$\theta = 63.7777\ldots^\circ$$

The common region is made from two minor segments.
For the circle with centre O:

$$\text{segment area} = \frac{50}{360}\pi(5)^2 - \frac{1}{2}(5)^2\sin 50^\circ = 1.3328\ldots$$

For the circle with centre P:

$$\text{segment area} = \frac{\theta}{360}\pi(4)^2 - \frac{1}{2}(4)^2\sin\theta$$

$$= 1.7284\ldots$$

Total area of region R:

$$1.3328\ldots + 1.7284\ldots = 3.0611\ldots$$

FINAL ANSWER

$\boxed{3.06 \text{ cm}^2}$

CLASSIFIED PROBLEM Paper: P1H Session: May 2023 Q#: 22 Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: May 2023 P1H

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

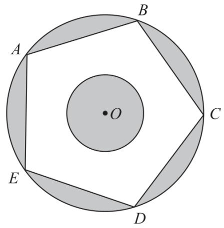


Diagram **NOT** accurately drawn

A, B, C, D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

shaded area = unshaded area

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1H | Question 22

Circles, Arcs & Sectors

Topic check. Moved to Circles, Arcs & Sectors.

Method

The regular pentagon has side length 8 cm.
For a regular pentagon,

$$R = \frac{s}{2\sin 36^\circ}$$

where R is the radius of the larger circle.

$$R = \frac{8}{2\sin 36^\circ} = 6.8052\ldots$$

Area of the larger circle:

$$\pi R^2 = 145.4898\ldots$$

Area of the regular pentagon:

$$\frac{5s^2}{4\tan 36^\circ} = \frac{5(8)^2}{4\tan 36^\circ} = 110.1106\ldots$$

Let the radius of the smaller circle be r.
The shaded area is:

$$\text{larger circle area} - \text{pentagon area} + \pi r^2$$

The unshaded area is:

$$\text{pentagon area} - \pi r^2$$

These are equal, so

$$145.4898 - 110.1106 + \pi r^2 = 110.1106 - \pi r^2$$

$$2\pi r^2 = 2(110.1106) - 145.4898$$

$$\pi r^2 = 37.3657\ldots$$

$$r = \sqrt{\frac{37.3657\ldots}{\pi}} = 3.4487\ldots$$

FINAL ANSWER

$\boxed{3.45 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P1H **Session: May 2024** **Q#: 20** **Marks: 4**

TOPIC: **Circles, Arcs & Sectors**

Source: May 2024 P1H

20 The diagram shows a sector *OABC* of a circle centre *O*

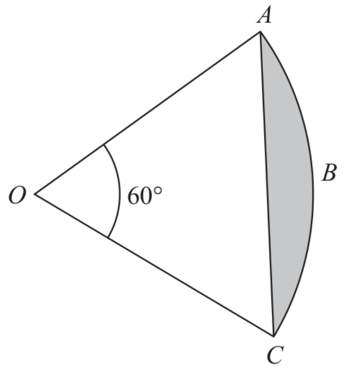


Diagram **NOT** accurately drawn

Angle $AOC = 60°$

The area of the shaded segment *ABC* is $38\,\text{cm}^2$

Work out the perimeter of the shaded segment *ABC*
Give your answer correct to one decimal place.

.... cm

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 20

Circles, Arcs & Sectors

Topic check. Correct.

Method

Let the radius of the sector be r.
The shaded segment is

$$\text{sector area} - \text{triangle area}$$

The angle is $60°$, so

$$38 = \frac{60}{360}\pi r^2 - \frac{1}{2}r^2 \sin 60°$$

$$38 = r^2\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right)$$

$$r = 20.4815\ldots$$

The perimeter of the shaded segment is the arc AC plus the chord AC.
Arc length:

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

Chord length:

$$2r \sin 30° = r$$

So the perimeter is

$$\frac{\pi r}{3} + r$$

$$= \frac{\pi(20.4815\ldots)}{3} + 20.4815\ldots$$

$$= 41.9296\ldots$$

FINAL ANSWER

$\boxed{41.9\text{ cm}}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2023 | Q#: 20 | Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Source: Nov 2023 P2H

20 The diagram shows equilateral triangle ABC with sides of length 10 cm.
A circle is drawn inside the triangle.

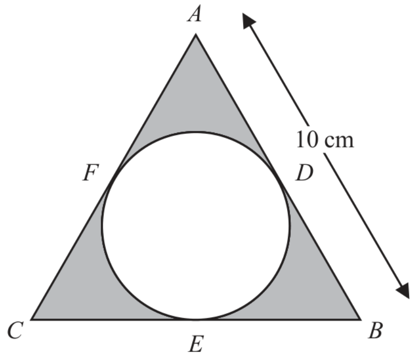


Diagram **NOT** accurately drawn

D, E and F are points on the circle.

ADB, BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | Question 20

Circles, Arcs & Sectors

Topic check. Moved to Circles, Arcs & Sectors.

Method

The triangle is equilateral with side length 10 cm.

Area of the triangle:

$$\frac{\sqrt{3}}{4}(10)^2 = 25\sqrt{3}$$

For an equilateral triangle, the inradius is

$$r = \frac{s\sqrt{3}}{6}$$

So

$$r = \frac{10\sqrt{3}}{6} = \frac{5\sqrt{3}}{3}$$

Area of the circle:

$$\pi r^2 = \pi\left(\frac{5\sqrt{3}}{3}\right)^2 = \frac{25\pi}{3}$$

The shaded area is

$$25\sqrt{3} - \frac{25\pi}{3} = 17.1213\ldots$$

FINAL ANSWER

$\boxed{17.1\ \text{cm}^2}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2024 | Q#: 25 | Marks: 5

TOPIC: Circles, Arcs & Sectors

Source: Nov 2024 P2H

25 The diagram shows an equilateral triangle ABC and a circle with centre O

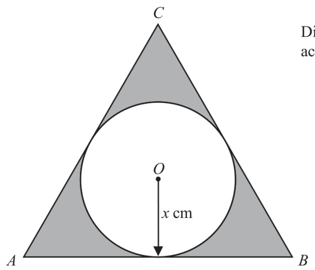


Diagram **NOT** accurately drawn

AB, BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2, of the regions shown shaded in the diagram is nx^2

Find the value of n
Give your answer correct to 3 significant figures.

n = ...

(Total for Question 25 is 5 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Nov 2024 P2H | Question 25
Circles, Arcs & Sectors

Topic check. Moved to Circles, Arcs & Sectors.

Method
The circle has radius x, so the inradius of the equilateral triangle is x.
For an equilateral triangle with side length a,

$$\text{inradius} = \frac{a\sqrt{3}}{6}$$

So

$$x = \frac{a\sqrt{3}}{6}$$

$$a = \frac{6x}{\sqrt{3}} = 2\sqrt{3}x$$

Area of the triangle:

$$\frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}(2\sqrt{3}x)^2$$

$$= \frac{\sqrt{3}}{4}(12x^2) = 3\sqrt{3}x^2$$

Area of the circle:

$$\pi x^2$$

The shaded area is

$$3\sqrt{3}x^2 - \pi x^2$$

$$= (3\sqrt{3} - \pi)x^2$$

Therefore

$$n = 3\sqrt{3} - \pi$$

FINAL ANSWER
$\boxed{3\sqrt{3} - \pi}$

Topic 16

Sine, Cosine Rule & Area of Triangles

CLASSIFIED PROBLEM Paper: 4WM1H Session: May2025 Q#: 21 Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: May 2025 4WM1H

21

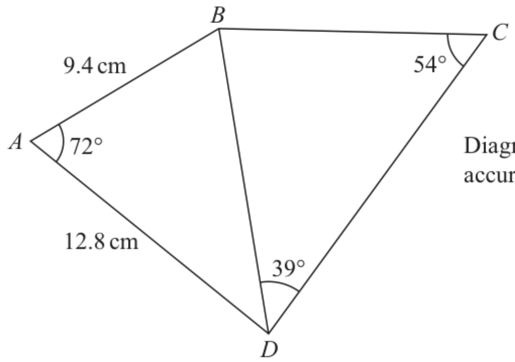


Diagram **NOT** accurately drawn

Work out the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 21
Sine, Cosine Rule & Area of Triangles

Topic check. Was tagged Algebra Toolkit; actually Sine & Cosine Rules.

Solution

First find BD in triangle ABD using the cosine rule.

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ$$

$$BD = 13.3356\ldots$$

In triangle BCD, use the sine rule.

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ}$$

$$BC = 13.3356\ldots \times \frac{\sin 39^\circ}{\sin 54^\circ}$$

$$BC = 10.3735\ldots$$

To 3 significant figures,

$$BC = 10.4$$

FINAL ANSWER
10.4 cm

Topic 17

3D Pythagoras & Trigonometry

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: May2025 | Q#: 23 | Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

Source: May 2025 4WM1H

23 The diagram shows cuboid *ABCDEFGH*

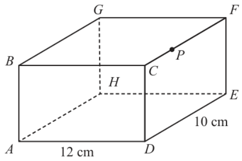


Diagram **NOT** accurately drawn

The point P lies on CF

$AD = 12$ cm $\quad DE = 10$ cm $\quad CP = 3$ cm

The angle of elevation of P from A is $24°$

Calculate the size of angle APG
Give your answer correct to the nearest degree.
Show your working clearly.

.... °

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 23

3D Pythagoras & Trigonometry

Topic check. Was tagged Algebra Toolkit; actually 3D Pythagoras & Trigonometry.

Solution

Point P is 3 cm from C along CF. The horizontal distance from A to the point directly below P is

$$\sqrt{12^2 + 3^2} = \sqrt{153}$$

Using the angle of elevation,

$$h = \sqrt{153}\tan 24^\circ = 5.507\ldots$$

So

$$AP = \frac{\sqrt{153}}{\cos 24^\circ} = 13.5399\ldots$$

On the top face,

$$PG = \sqrt{12^2 + 7^2} = \sqrt{193} = 13.8924\ldots$$

Also,

$$AG = \sqrt{12^2 + 10^2 + h^2} = 16.561\ldots$$

Use the cosine rule in triangle APG.

$$\cos\angle APG = \frac{AP^2 + PG^2 - AG^2}{2(AP)(PG)}$$

$$\angle APG = 49.163\ldots^\circ$$

To the nearest degree,

$$49^\circ$$

FINAL ANSWER

49°

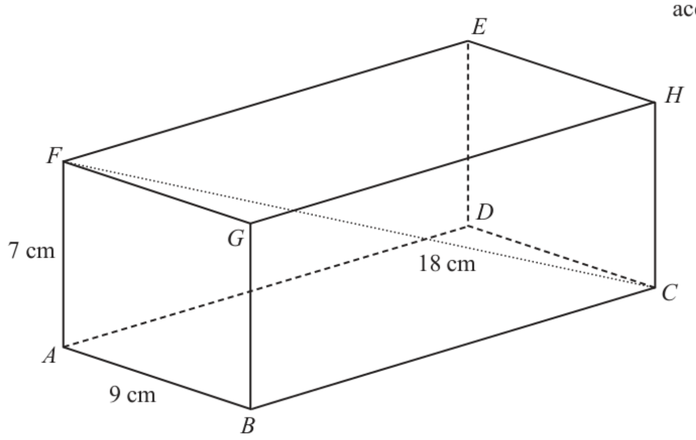
CLASSIFIED PROBLEM Paper: 4WM1HR Session: May2025 Q#: 20 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Source: May 2025 4WM1HR

20 The diagram shows cuboid *ABCDEFGH*

Diagram **NOT** accurately drawn



$AB = 9\,\text{cm}$ $AF = 7\,\text{cm}$ $FC = 18\,\text{cm}$

Calculate the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 20
3D Pythagoras & Trigonometry

Topic check. Correct.

Method

In the rectangle $ABCF$, use Pythagoras with the diagonal FC:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

Now use Pythagoras in triangle ABC:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

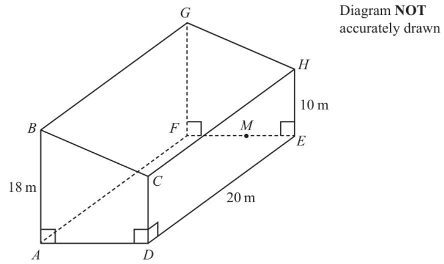
FINAL ANSWER
13.9 cm to 3 significant figures

CLASSIFIED PROBLEM Paper: 4WM1H Session: Nov2025 Q#: 26 Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Source: Nov 2025 4WM1H

26 The diagram shows prism $ABCDEFGH$



$AB = FG = 18\text{ m}$ $DC = EH = 10\text{ m}$ $AD = FE$

$DE = CH = AF = BG = 20\text{ m}$

angle BAD = angle ADC = angle CDE = 90°

M is the point on FE such that $FM = \frac{3}{5}FE$

Given that angle $GMF = 60°$

calculate the angle of elevation of G from D
Give your answer correct to one decimal place.
Show your working clearly.

..................................°

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 26
3D Pythagoras & Trigonometry

Solution

From triangle GMF,

$$\tan 60^\circ = \frac{FG}{FM} = \frac{18}{FM}$$

$$FM = \frac{18}{\tan 60^\circ} = 6\sqrt{3}$$

Since $FM = \frac{3}{5}FE$,

$$FE = 10\sqrt{3}$$

The horizontal distance from D to the point below G is

$$\sqrt{20^2 + (10\sqrt{3})^2} = 10\sqrt{7}$$

If θ is the angle of elevation,

$$\tan\theta = \frac{18}{10\sqrt{7}}$$

$$\theta \approx 34.2289^\circ$$

FINAL ANSWER

34.2°.

Topic 18

Sine, Cosine Rule & Area of Triangles

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jan 2020 | Q#: 21 | Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: Jan 2020 P1HR

21 The diagram shows the prism *ABCDEF* with cross section triangle *ABC*.

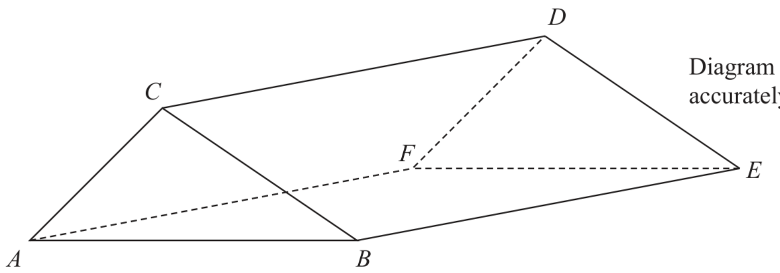


Diagram **NOT** accurately drawn

Angle $BEC = 40°$ and angle ACB is obtuse.
$AC = 6$ cm and $CE = 13$ cm

The area of triangle ABC is 22 cm^2

Calculate the length of AB.
Give your answer correct to one decimal place.

.... cm

(Total for Question 21 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1HR | Question 21

Sine, Cosine Rule & Area of Triangles

Topic check. Correct.

Method

In triangle BEC, $CE = 13$ and $\angle BEC = 40°$.
Since this is a prism, BE is perpendicular to the cross section, so triangle BEC is right-angled at B.

$$BC = 13 \sin 40°$$

$$BC = 8.356238\ldots$$

The area of triangle ABC is 22 cm^2.
Using

$$\text{Area} = \frac{1}{2}ab\sin C$$

$$22 = \frac{1}{2}(AC)(BC)\sin(\angle ACB)$$

$$22 = \frac{1}{2}(6)(8.356238\ldots)\sin(\angle ACB)$$

$$\sin(\angle ACB) = 0.877587\ldots$$

The angle ACB is obtuse, so

$$\angle ACB = 180° - \sin^{-1}(0.877587\ldots)$$

$$\angle ACB = 118.647\ldots°$$

Use the cosine rule:

$$AB^2 = AC^2 + BC^2 - 2(AC)(BC)\cos(\angle ACB)$$

$$AB^2 = 6^2 + (8.356238\ldots)^2 - 2(6)(8.356238\ldots)\cos(118.647\ldots°)$$

$$AB = 12.4056\ldots$$

Correct to one decimal place:

$$12.4$$

FINAL ANSWER

$\boxed{12.4 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2022 | Q#: 23 | Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: Jan 2022 P2H

23 The diagram shows triangle PQR

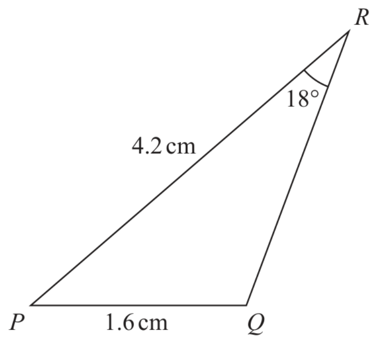


Diagram **NOT** accurately drawn

$PQ = 1.6$ cm $\quad$ $PR = 4.2$ cm $\quad$ Angle $PRQ = 18°$

Given that angle PQR is obtuse,

work out the area of triangle PQR
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 23
Sine, Cosine Rule & Area of Triangles

Topic check. Correct.

Method

In triangle PQR,

$$PQ = 1.6, \qquad PR = 4.2, \qquad \angle PRQ = 18^\circ$$

Use the sine rule:

$$\frac{\sin Q}{PR} = \frac{\sin R}{PQ}$$

$$\frac{\sin Q}{4.2} = \frac{\sin 18^\circ}{1.6}$$

$$\sin Q = \frac{4.2 \sin 18^\circ}{1.6}$$

$$\sin Q = 0.811169\ldots$$

Since angle PQR is obtuse,

$$Q = 180^\circ - \sin^{-1}(0.811169\ldots)$$

$$Q = 125.7896\ldots^\circ$$

Now find angle P:

$$P = 180^\circ - 18^\circ - 125.7896\ldots^\circ$$

$$P = 36.2103\ldots^\circ$$

Area of triangle PQR:

$$\frac{1}{2}(PQ)(PR)\sin P$$

$$= \frac{1}{2}(1.6)(4.2)\sin(36.2103\ldots^\circ)$$

$$= 1.984925\ldots$$

Correct to 3 significant figures:

$$1.98$$

FINAL ANSWER

$\boxed{1.98\ \text{cm}^2}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2023 | Q#: 22 | Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: Jun 2023 P2H

22 The diagram shows a triangle *ABC* and a flagpole *BF*

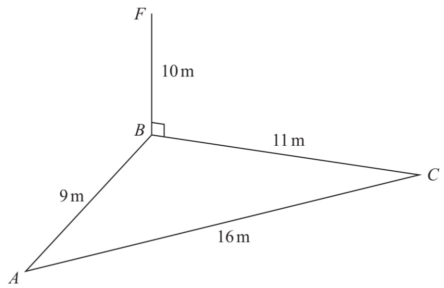


Diagram **NOT** accurately drawn

A, *B* and *C* are points on horizontal ground.

BF is vertical.

$AB = 9\text{ m}$ $\quad BC = 11\text{ m}$ $\quad AC = 16\text{ m}$ $\quad BF = 10\text{ m}$

D is the point on *AC* such that angle $BDC = 90°$

Work out the size of the angle of elevation of the point *F* from the point *D*
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 22

Sine, Cosine Rule & Area of Triangles

Topic check. Moved to Sine, Cosine Rule & Area of Triangles.

Method

In triangle ABC,

$$AB = 9, \qquad BC = 11, \qquad AC = 16$$

Use Heron's formula.

$$s = \frac{9 + 11 + 16}{2} = 18$$

$$\text{area} = \sqrt{18(18-9)(18-11)(18-16)}$$

$$= \sqrt{18 \cdot 9 \cdot 7 \cdot 2} = 47.6235\ldots$$

D is the foot of the perpendicular from B to AC, so BD is the height of triangle ABC.

$$\text{area} = \frac{1}{2} \times AC \times BD$$

$$47.6235\ldots = \frac{1}{2} \times 16 \times BD$$

$$BD = 5.9529\ldots$$

In right-angled triangle BDF,

$$\tan\theta = \frac{BF}{BD}$$

$$\tan\theta = \frac{10}{5.9529\ldots}$$

$$\theta = \tan^{-1}\left(\frac{10}{5.9529\ldots}\right) = 59.2349\ldots^\circ$$

FINAL ANSWER

$\boxed{59.2^\circ}$

CLASSIFIED PROBLEM Paper: P1H | Session: Jun 2025 | Q#: 20 | Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: Jun 2025 P1H

20

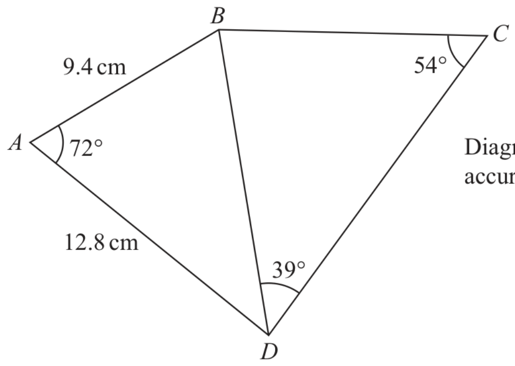


Diagram **NOT** accurately drawn

Work out the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 20
Sine, Cosine Rule & Area of Triangles

Topic check. Moved from Standard & Compound Units to Sine, Cosine Rule & Area of Triangles.

Method

First use triangle ABD.

$$AB = 9.4, \quad AD = 12.8, \quad \angle BAD = 72^\circ$$

Use the cosine rule to find BD:

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ$$

$$BD = 13.3356\ldots$$

In triangle BCD,

$$\angle BDC = 39^\circ, \quad \angle BCD = 54^\circ$$

Use the sine rule:

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ}$$

$$BC = \frac{13.3356\ldots \sin 39^\circ}{\sin 54^\circ}$$

$$BC = 10.3735\ldots$$

Correct to 3 significant figures,

$$BC = 10.4$$

FINAL ANSWER
10.4 cm

CLASSIFIED PROBLEM

Paper: P1HR | Session: May 2023 | Q#: 25 | Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: May 2023 P1HR

25 Here is a triangle *ABC*

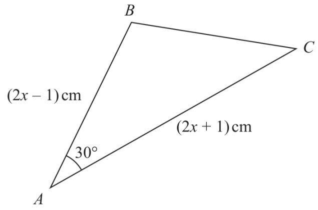


Diagram **NOT** accurately drawn

The area of the triangle is $(x^2 + x - 3.75)\,\text{cm}^2$

Find the size of the largest angle in triangle *ABC*
Give your answer correct to the nearest degree.

.... °

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 25

Sine, Cosine Rule & Area of Triangles

Topic check. Moved to Sine, Cosine Rule & Area of Triangles.

Method

The two sides around the 30° angle are

$$2x - 1 \quad \text{and} \quad 2x + 1$$

Using the area formula,

$$\text{area} = \frac{1}{2}ab\sin C$$

$$x^2 + x - 3.75 = \frac{1}{2}(2x-1)(2x+1)\sin 30°$$

Since $\sin 30° = \frac{1}{2}$,

$$x^2 + x - 3.75 = \frac{1}{4}(2x-1)(2x+1)$$

$$x^2 + x - 3.75 = \frac{1}{4}(4x^2 - 1)$$

$$x^2 + x - 3.75 = x^2 - 0.25$$

$$x = 3.5$$

So

$$AB = 2x - 1 = 6, \qquad AC = 2x + 1 = 8$$

Use the cosine rule to find BC:

$$BC^2 = 6^2 + 8^2 - 2(6)(8)\cos 30°$$

$$BC = 4.1063\ldots$$

The largest side is $AC = 8$, so the largest angle is $\angle B$.
Using the cosine rule again:

$$\cos B = \frac{6^2 + BC^2 - 8^2}{2(6)(BC)}$$

$$B = 103.0643\ldots°$$

FINAL ANSWER

$\boxed{103°}$

Topic 19

3D Pythagoras & Trigonometry

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2021 | Q#: 22 | Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jan 2021 P2HR

22 ABC is an isosceles triangle in a horizontal plane.
The point T is vertically above B.

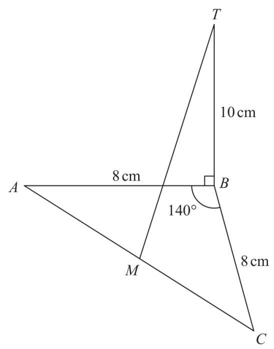


Diagram **NOT** accurately drawn

Angle $ABC = 140°$
$AB = BC = 8$ cm
$TB = 10$ cm
M is the midpoint of AC.

Calculate the size of the angle between MT and the horizontal plane ABC.
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2HR | Question 22

3D Pythagoras & Trigonometry

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

Since $AB = BC = 8$, triangle ABC is isosceles.
The line from B to the midpoint M bisects angle ABC.

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

In right triangle ABM,

$$\cos 70^\circ = \frac{BM}{8}$$

$$BM = 8\cos 70^\circ$$

Since T is vertically above B, the projection of MT onto the horizontal plane is MB.
Let the required angle be θ.

$$\tan\theta = \frac{TB}{BM}$$

$$\tan\theta = \frac{10}{8\cos 70^\circ}$$

$$\theta = 74.7^\circ$$

FINAL ANSWER

74.7°

CLASSIFIED PROBLEM

Paper: P1H | Session: Jan 2022 | Q#: 21 | Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jan 2022 P1H

21 The diagram shows the prism *ABCDEFGHJK* with horizontal base *AEFG*

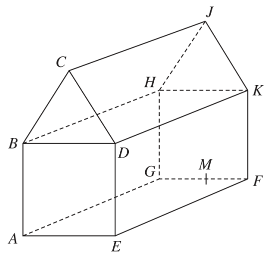


Diagram **NOT** accurately drawn

ABCDE is a cross section of the prism where
ABDE is a square
BCD is an equilateral triangle

$EF = 2 \times AE$
M is the midpoint of *GF* so that *JM* is vertical.

Angle $MAJ = y°$

Given that $\tan y° = T$

find the value of *T*, giving your answer in the form $\dfrac{\sqrt{p} + \sqrt{q}}{17}$ where *p* and *q* are integers.

T = ...

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 21

3D Pythagoras & Trigonometry

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

Let

$$AE = x$$

Since $ABDE$ is a square,

$$AB = BD = DE = AE = x$$

Since BCD is an equilateral triangle, the height from C to BD is

$$\frac{\sqrt{3}}{2}x$$

Therefore the vertical height from the base to J is

$$JM = x + \frac{\sqrt{3}}{2}x$$

$$JM = \frac{x(2+\sqrt{3})}{2}$$

Also,

$$EF = 2AE = 2x$$

Since M is the midpoint of GF, the horizontal distance from A to M is

$$AM = \sqrt{\left(\frac{x}{2}\right)^2 + (2x)^2}$$

$$AM = \sqrt{\frac{x^2}{4} + 4x^2}$$

$$AM = \frac{x\sqrt{17}}{2}$$

In right triangle AMJ,

$$\tan y^\circ = \frac{JM}{AM}$$

$$T = \frac{\frac{x(2+\sqrt{3})}{2}}{\frac{x\sqrt{17}}{2}}$$

$$T = \frac{2+\sqrt{3}}{\sqrt{17}}$$

Rationalise the denominator:

$$T = \frac{(2+\sqrt{3})\sqrt{17}}{17}$$

$$T = \frac{2\sqrt{17}+\sqrt{51}}{17}$$

$$T = \frac{\sqrt{68}+\sqrt{51}}{17}$$

FINAL ANSWER

$T = \dfrac{\sqrt{68}+\sqrt{51}}{17}$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2023 | Q#: 22 | Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jun 2023 P2HR

22 Here is a cuboid *ABCDEFGH*

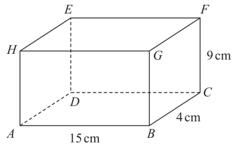


Diagram **NOT** accurately drawn

$AB = 15$ cm $BC = 4$ cm $CF = 9$ cm

(a) Work out the length of *BE*
Give your answer correct to 3 significant figures.

.... cm
(2)

Here is a cuboid *PQRSTUVW*

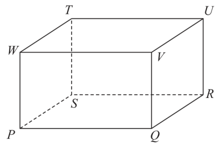


Diagram **NOT** accurately drawn

$PR = 42$ cm

The size of the angle between *PU* and the plane *PQRS* is 30°

M is the midpoint of *PR*

(b) Work out the size of angle *UMR*
Give your answer correct to 3 significant figures.

.... °
(3)

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 22

3D Pythagoras & Trigonometry

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

For part (a), BE is the space diagonal using the three perpendicular lengths:

$$AB = 15, \qquad BC = 4, \qquad CF = 9$$

$$BE^2 = 15^2 + 4^2 + 9^2$$

$$BE^2 = 225 + 16 + 81 = 322$$

$$BE = \sqrt{322} = 17.9 \text{ cm}$$

For part (b), the projection of PU onto the plane $PQRS$ is PR.

$$PR = 42$$

The angle between PU and the plane is $30°$, so

$$\tan 30° = \frac{RU}{PR}$$

$$RU = 42 \tan 30°$$

Since M is the midpoint of PR,

$$MR = 21$$

In right triangle MRU,

$$\tan \angle UMR = \frac{RU}{MR}$$

$$\tan \angle UMR = \frac{42 \tan 30°}{21}$$

$$\tan \angle UMR = 2 \tan 30°$$

$$\angle UMR = 49.1°$$

FINAL ANSWER

$BE = 17.9$ cm, $\angle UMR = 49.1°$

CLASSIFIED PROBLEM

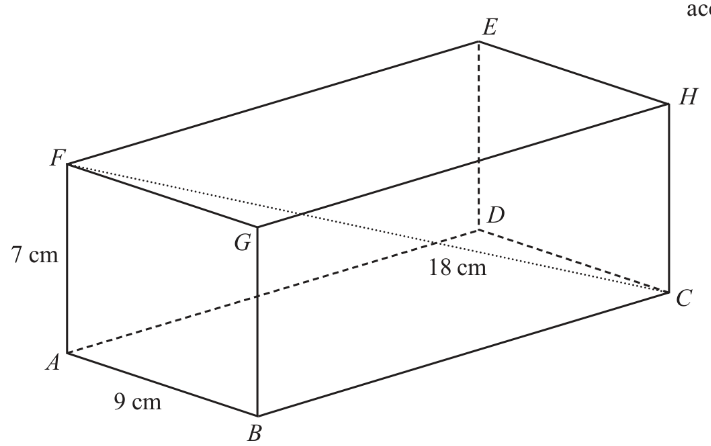
Paper: P1HR | Session: Jun 2025 | Q#: 20 | Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jun 2025 P1HR

20 The diagram shows cuboid *ABCDEFGH*

Diagram **NOT** accurately drawn



$AB = 9\,\text{cm}$ $\quad AF = 7\,\text{cm}$ $\quad FC = 18\,\text{cm}$

Calculate the length of BC
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 20

3D Pythagoras & Trigonometry

Topic check. Correct.

Method

In the rectangle $ABCF$, use Pythagoras with the diagonal FC:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

Now use Pythagoras in triangle ABC:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

FINAL ANSWER

13.9 cm to 3 significant figures

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2025 | Q#: 24 | Marks: 4

Topic: **3D Pythagoras & Trigonometry**

Source: Jun 2025 P2H

24 The diagram shows a square-based pyramid *ABCDE*

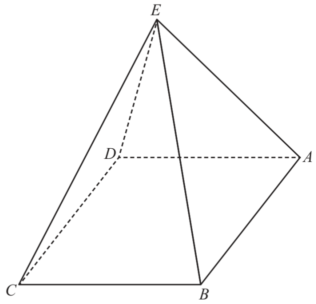


Diagram **NOT** accurately drawn

$EA = EB = EC = ED$

M is the centre of the horizontal square base $ABCD$
Q is the midpoint of AB
Angle $EQM = 80°$

$EA : AB = n : 1$

Find the value of n
Give your answer correct to 3 significant figures.

$n =$ ....

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 24
3D Pythagoras & Trigonometry

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

Let

$$AB = s$$

Since M is the centre of the square base and Q is the midpoint of AB,

$$QM = \frac{s}{2}$$

Since $EA = EB = EC = ED$, the point E is vertically above M.
In right triangle EQM,

$$\tan 80^\circ = \frac{EM}{QM}$$

$$EM = \frac{s}{2} \tan 80^\circ$$

The distance from the centre of a square to a vertex is half the diagonal:

$$AM = \frac{s\sqrt{2}}{2}$$

In right triangle EMA,

$$EA^2 = EM^2 + AM^2$$

$$EA^2 = \left(\frac{s}{2} \tan 80^\circ\right)^2 + \left(\frac{s\sqrt{2}}{2}\right)^2$$

$$EA = s\sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

So

$$\frac{EA}{AB} = \sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

$$n = 2.92$$

FINAL ANSWER
$n = 2.92$

CLASSIFIED PROBLEM Paper: P2HR Session: Jun 2025 Q#: 23 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jun 2025 P2HR

23 The diagram shows a triangular prism, *ABCDEF*, with a horizontal rectangular base *ABCD*

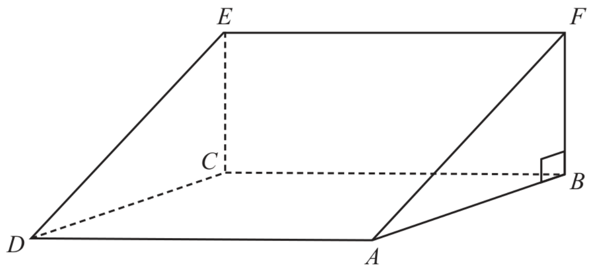


Diagram **NOT** accurately drawn

M is the midpoint of the line *AC*

$AC = 40\,\text{cm}$ angle $CAE = 35°$ angle $ABF = 90°$

Work out the size of angle *CME*
Give your answer correct to 3 significant figures.

.... °

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 23
3D Pythagoras & Trigonometry

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

$$AC = 40$$

The point E is vertically above C, so triangle ACE is right-angled at C.

$$\angle CAE = 35^\circ$$

$$\tan 35^\circ = \frac{CE}{AC}$$

$$CE = 40 \tan 35^\circ$$

Since M is the midpoint of AC,

$$CM = 20$$

In right triangle CME,

$$\tan \angle CME = \frac{CE}{CM}$$

$$\tan \angle CME = \frac{40 \tan 35^\circ}{20}$$

$$\tan \angle CME = 2 \tan 35^\circ$$

$$\angle CME = 54.5^\circ$$

FINAL ANSWER
54.5°

CLASSIFIED PROBLEM

Paper: P1H | Session: May 2021 | Q#: 22 | Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Source: May 2021 P1H

22 The diagram shows a triangular prism *ABCDEF* with a horizontal base *ABEF*.

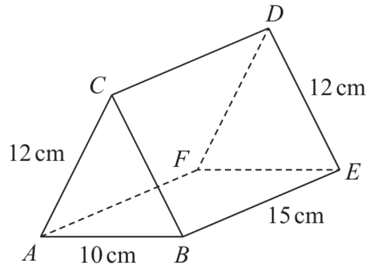


Diagram **NOT** accurately drawn

$AC = BC = FD = ED = 12\,\text{cm}$ $\quad AB = 10\,\text{cm}$ $\quad BE = 15\,\text{cm}$

Calculate the size of the angle between *AD* and the base *ABEF*.
Give your answer correct to 3 significant figures.

.... °

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 22
3D Pythagoras & Trigonometry

Topic check. Correct.

Method

The triangular end is isosceles, with

$$AC = BC = 12, \qquad AB = 10$$

The perpendicular height of the triangle is found by halving AB:

$$AM = 5$$

$$CM^2 = 12^2 - 5^2 = 144 - 25 = 119$$

$$CM = \sqrt{119}$$

The projection of AD on the horizontal base goes from A to the midpoint of FE. Its length is

$$\sqrt{15^2 + 5^2} = \sqrt{250}$$

So, for the angle θ between AD and the base,

$$\tan\theta = \frac{\sqrt{119}}{\sqrt{250}}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{119}}{\sqrt{250}}\right) = 34.602\ldots$$

FINAL ANSWER
$34.6°$

CLASSIFIED PROBLEM

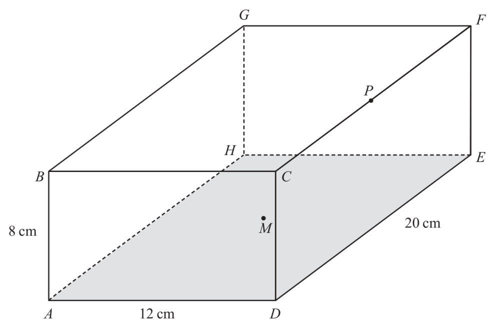
Paper: P1HR | Session: May 2024 | Q#: 22 | Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

Source: May 2024 P1HR

22 The diagram shows a cuboid *ABCDEFGH* with horizontal base *ADEH*

Diagram **NOT** accurately drawn



AB = 8 cm *AD* = 12 cm *DE* = 20 cm

M is the midpoint of the base *ADEH* and *P* is the midpoint of the edge *CF*

Work out the size of angle *BMP*
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1HR | Question 22

3D Pythagoras & Trigonometry

Topic check. Moved from Coordinate Geometry to 3D Pythagoras & Trigonometry.

Method

Use coordinates with

$$A = (0, 0, 0)$$

Since

$$AD = 12, \qquad DE = 20, \qquad AB = 8$$

take

$$B = (0, 0, 8)$$

The centre of the rectangular base $ADEH$ is

$$M = (6, 10, 0)$$

Point P is the midpoint of CF.

$$C = (12, 0, 8), \qquad F = (12, 20, 8)$$

$$P = (12, 10, 8)$$

Now

$$\vec{MB} = B - M = (-6, -10, 8)$$

$$\vec{MP} = P - M = (6, 0, 8)$$

Use the scalar product formula:

$$\vec{MB} \cdot \vec{MP} = (-6)(6) + (-10)(0) + (8)(8)$$

$$\vec{MB} \cdot \vec{MP} = 28$$

$$|\vec{MB}| = \sqrt{(-6)^2 + (-10)^2 + 8^2} = \sqrt{200}$$

$$|\vec{MP}| = \sqrt{6^2 + 0^2 + 8^2} = 10$$

$$\cos \angle BMP = \frac{28}{10\sqrt{200}}$$

$$\angle BMP = 78.6^\circ$$

FINAL ANSWER

78.6°

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2025 | Q#: 26 | Marks: 5

TOPIC: 3D Pythagoras & Trigonometry

Source: Nov 2025 P1H

26 Here is a prism *ABCDEFGH* with a horizontal, rectangular base *ABGF*

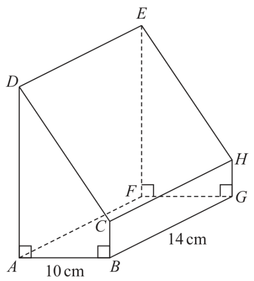


Diagram **NOT** accurately drawn

$AB = 10\,\text{cm}$ $\quad$ $BG = 14\,\text{cm}$

angle DAB = angle ABC = angle EFG = angle FGH = 90°

$BC = \frac{1}{5}AD$

The angle of elevation of *E* from *B* is 50°

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

.... cm^3

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 26

3D Pythagoras & Trigonometry

Topic check. Corrected from Right-Angled Triangles to 3D Pythagoras & Trigonometry.

Method

The horizontal base $ABGF$ is a rectangle.

$$AB = 10, \qquad BG = 14$$

So the horizontal distance from B to F is

$$BF = \sqrt{10^2 + 14^2} = \sqrt{296}$$

The angle of elevation of E from B is $50°$.
Since E is vertically above F,

$$\tan 50° = \frac{FE}{BF}$$

$$FE = \sqrt{296}\tan 50°$$

The prism has the same cross section throughout, so

$$AD = FE = \sqrt{296}\tan 50°$$

Also

$$BC = \frac{1}{5}AD$$

Area of the trapezium cross section $ABCD$:

$$\frac{1}{2}(AD + BC) \times AB$$

$$= \frac{1}{2}\left(AD + \frac{1}{5}AD\right) \times 10$$

$$= 6AD$$

Volume of the prism:

$$6AD \times 14 = 84AD$$

$$= 84\sqrt{296}\tan 50°$$

$$= 1722.311\ldots$$

Correct to 3 significant figures:

$$1720$$

FINAL ANSWER

$\boxed{1720 \text{ cm}^3}$

Topic 20

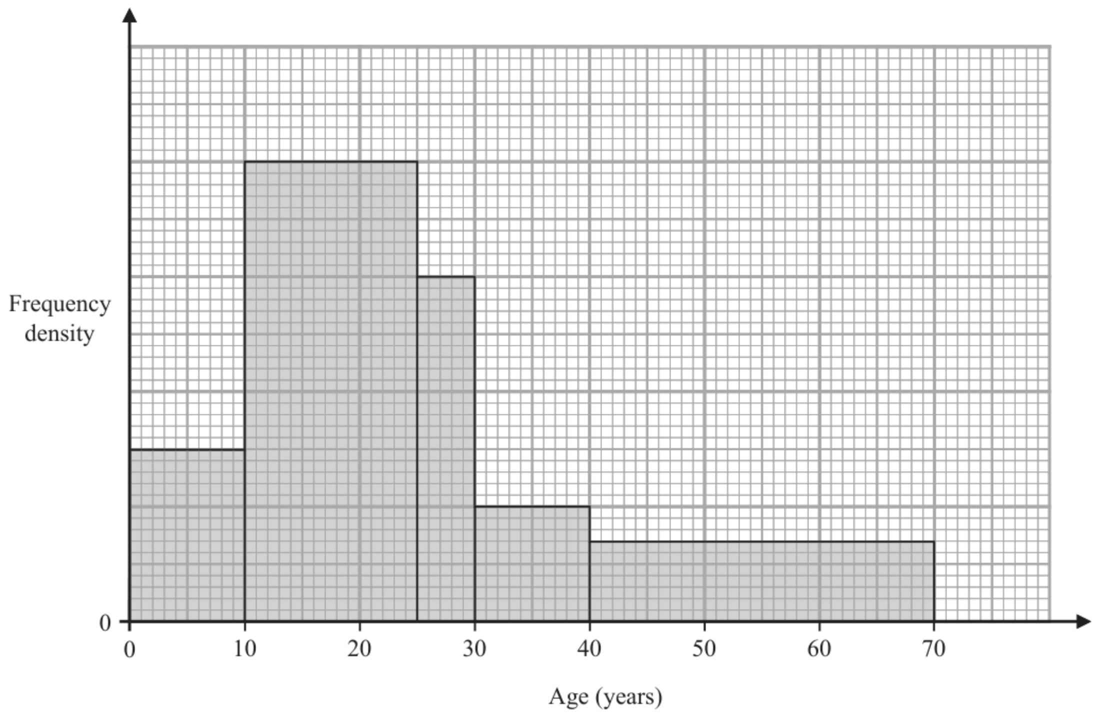
Histograms

CLASSIFIED PROBLEM Paper: 4WM1HR Session: May2025 Q#: 21 Marks: 3

TOPIC: **Histograms**

Source: May 2025 4WM1HR

21 The histogram shows information about the ages of the members of a chess club.



There are 60 members aged between 10 and 25
There are no members older than 70

Work out an estimate for the number of members who are over 35 years of age.

....

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 21

Histograms

Solution

The interval 10 to 25 has width 15 and frequency 60, so its frequency density is

$$\frac{60}{15} = 4$$

Using that scale from the histogram:

$$35 \text{ to } 40 : 5 \times 1 = 5$$

$$40 \text{ to } 70 : 30 \times 0.75 = 22.5$$

$$5 + 22.5 = 27.5$$

FINAL ANSWER

about 28 members.

CLASSIFIED PROBLEM Paper: 4WM1H Session: Nov2025 Q#: 20 Marks: 3

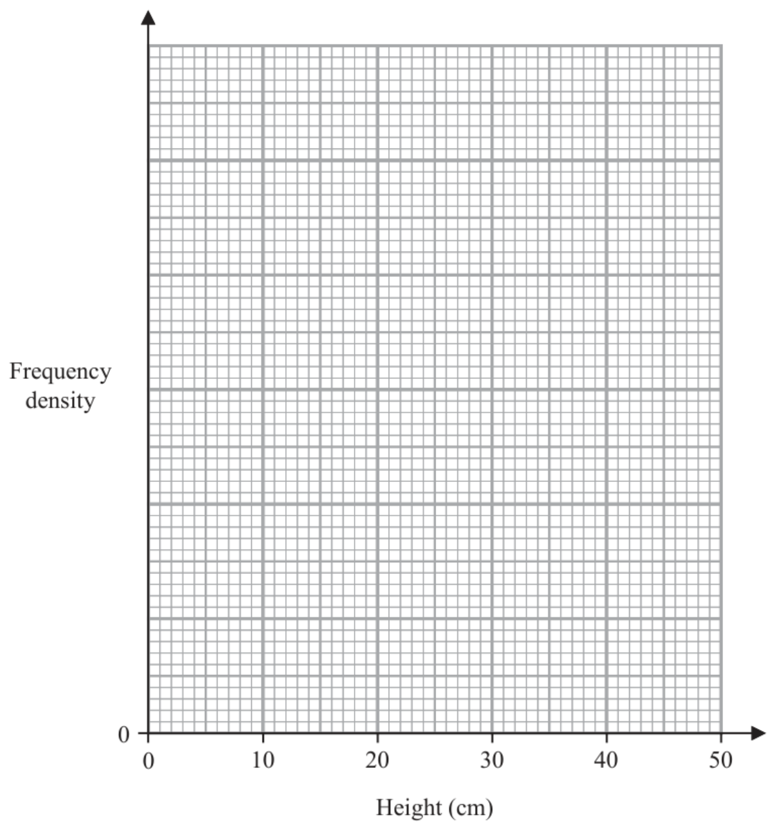
TOPIC: **Histograms**

Source: Nov 2025 4WM1H

20 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	**Frequency**
$0 < h \leqslant 5$	12
$5 < h \leqslant 15$	34
$15 < h \leqslant 30$	45
$30 < h \leqslant 50$	28

On the grid, draw a histogram for this information.
You must include a scale on the vertical axis.



(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 20

Histograms

Solution

Frequency density $= \dfrac{\text{frequency}}{\text{class width}}$.

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

$$5 < h \leq 15 : \frac{34}{10} = 3.4$$

$$15 < h \leq 30 : \frac{45}{15} = 3$$

$$30 < h \leq 50 : \frac{28}{20} = 1.4$$

FINAL ANSWER

draw bars with densities $2.4, 3.4, 3, 1.4$.

Topic 21

Probability Toolkit

CLASSIFIED PROBLEM Paper: 4WM1H Session: May2025 Q#: 22 Marks: 3

TOPIC: **Probability Toolkit**

Source: May 2025 4WM1H

22 A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

....

(Total for Question 22 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 22
Probability Toolkit

Topic check. Probability Toolkit. The tag is correct.

Solution

Total ways to choose 3 counters:

$$\binom{20}{3} = 1140$$

Exactly two red:

$$\binom{9}{2}\binom{11}{1} = 396$$

Exactly two yellow:

$$\binom{7}{2}\binom{13}{1} = 273$$

Exactly two green:

$$\binom{4}{2}\binom{16}{1} = 96$$

Favourable outcomes:

$$396 + 273 + 96 = 765$$

$$P = \frac{765}{1140} = \frac{51}{76}$$

FINAL ANSWER

$\frac{51}{76}$

Topic 22

Histograms

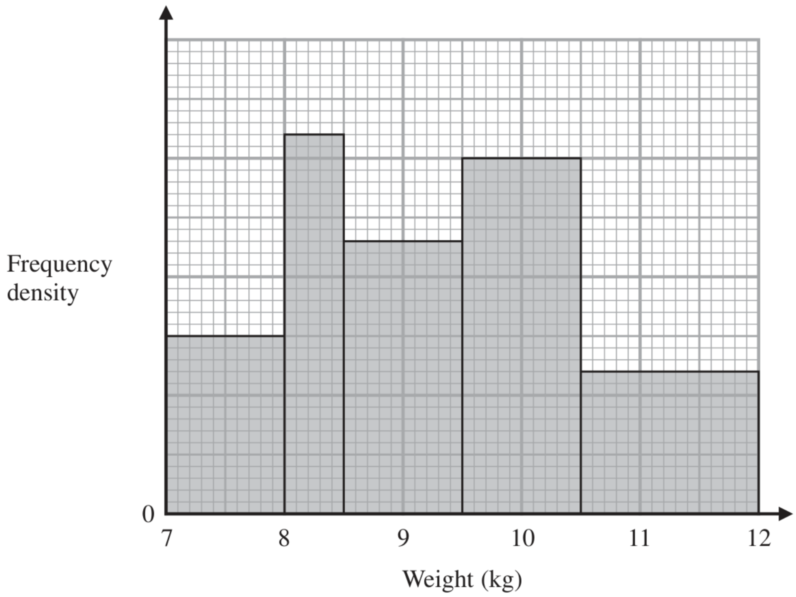
CLASSIFIED PROBLEM

Paper: P1HR | Session: Jan 2022 | Q#: 21 | Marks: 3

TOPIC: **Histograms**

Source: Jan 2022 P1HR

21



The histogram gives information about the weights, in kg, of all the watermelons in a field.

There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

....

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 21

Histograms

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $8 \leq w < 8.5$, the frequency is 16 and the class width is 0.5, so

$$\text{frequency density} = \frac{16}{0.5} = 32$$

Using this scale from the histogram, the frequency densities are:

$$15,\ 32,\ 23,\ 30,\ 12$$

The total frequency is

$$(1)(15) + (0.5)(32) + (1)(23) + (1)(30) + (1.5)(12)$$

$$= 15 + 16 + 23 + 30 + 18$$

$$= 102$$

FINAL ANSWER

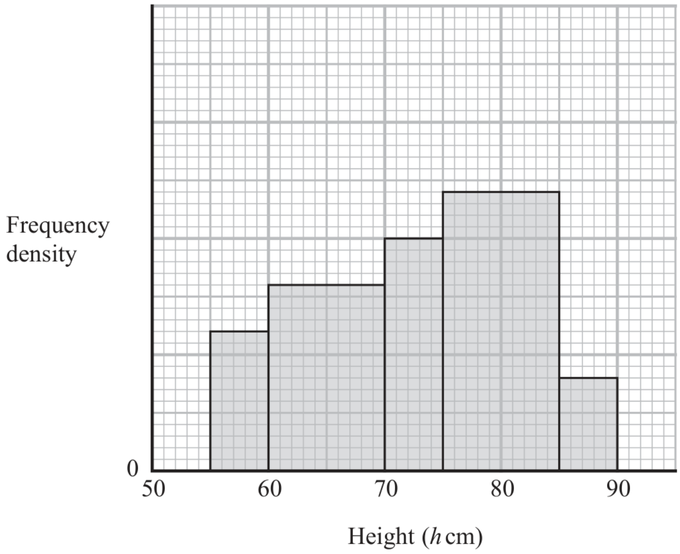
102 watermelons

CLASSIFIED PROBLEM Paper: P2H | Session: Jan 2022 | Q#: 20 | Marks: 4

Topic: **Histograms**

Source: Jan 2022 P2H

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leqslant 85$
One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

....

(Total for Question 20 is 4 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jan 2022 P2H | Question 20
Histograms

Topic check. Correct.

Method
For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $75 < h \leq 85$, the frequency is 12 and the class width is 10, so

$$\text{frequency density} = \frac{12}{10} = 1.2$$

Using this scale from the histogram, the frequency densities are:

$$0.6,\ 0.8,\ 1.0,\ 1.2,\ 0.4$$

The total number of tomato plants is

$$(5)(0.6) + (10)(0.8) + (5)(1.0) + (10)(1.2) + (5)(0.4)$$

$$= 3 + 8 + 5 + 12 + 2 = 30$$

For $h > 82.5$:

$$(2.5)(1.2) + (5)(0.4) = 3 + 2 = 5$$

So the probability is

$$\frac{5}{30} = \frac{1}{6}$$

FINAL ANSWER
$\frac{1}{6}$

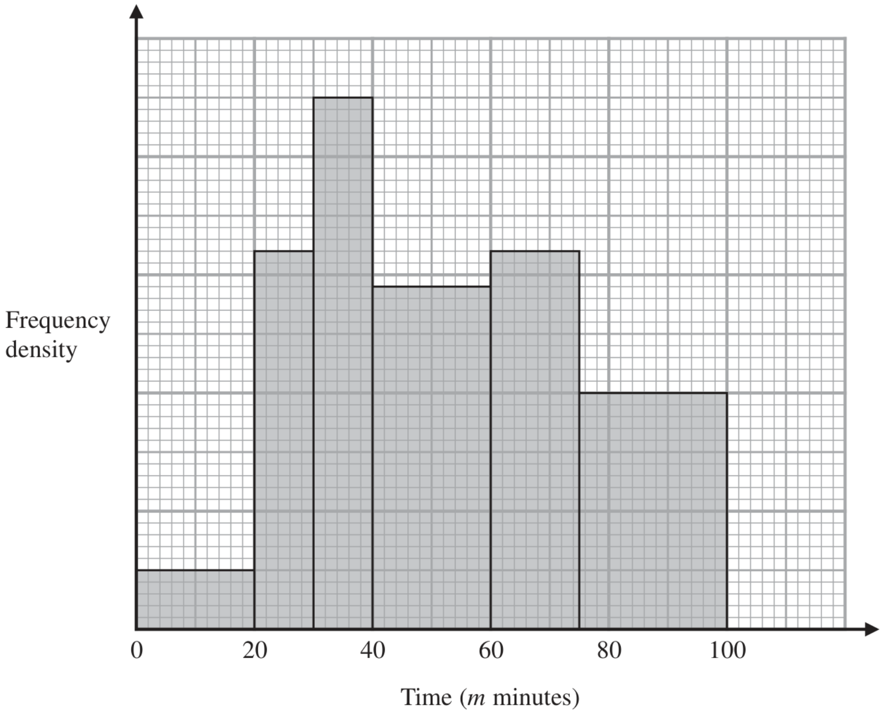
CLASSIFIED PROBLEM

Paper: P1H | Session: Jun 2022 | Q#: 21 | Marks: 3

TOPIC: **Histograms**

Source: Jun 2022 P1H

21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$
There are 10 children for whom $m \leqslant 20$

Work out an estimate for the number of children for whom $50 < m \leqslant 80$

....

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 21
Histograms

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $m \leq 20$, the frequency is 10 and the class width is 20, so

$$\text{frequency density} = \frac{10}{20} = 0.5$$

Using this scale from the histogram:

$$40 < m \leq 60 \quad \text{has frequency density } 2.9$$

$$60 < m \leq 75 \quad \text{has frequency density } 3.2$$

$$75 < m \leq 100 \quad \text{has frequency density } 2.0$$

For $50 < m \leq 80$, use part of each relevant bar:

$$(10)(2.9) + (15)(3.2) + (5)(2.0)$$

$$= 29 + 48 + 10$$

$$= 87$$

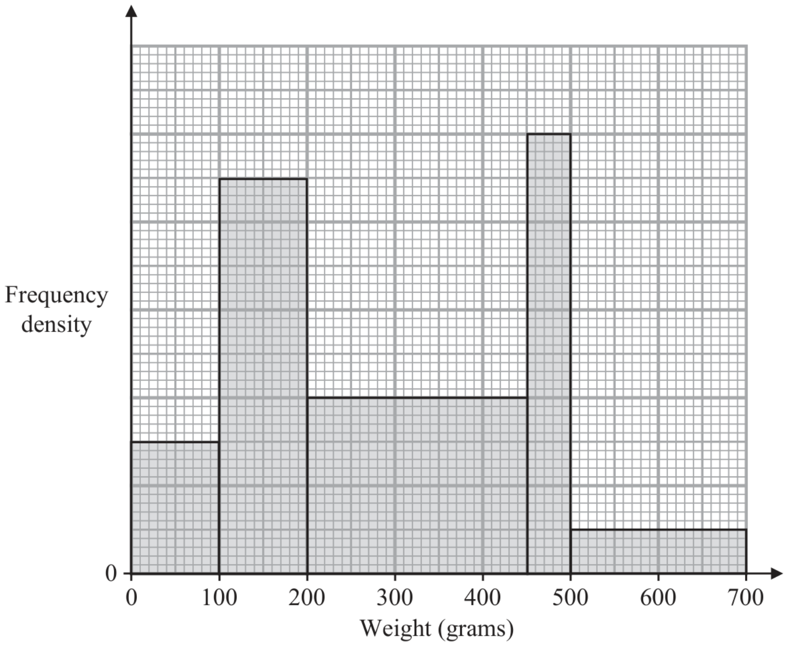
FINAL ANSWER
87 children

CLASSIFIED PROBLEM Paper: P2HR | Session: Jun 2025 | Q#: 20 | Marks: 4

TOPIC: **Histograms**

Source: Jun 2025 P2HR

20 The histogram gives some information about the weights, in grams, of some books.



75 books weigh less than 100 grams.

A book is chosen at random.

Find an estimate for the probability that this book weighs between 400 grams and 600 grams.

....

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 20
Histograms

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For books weighing less than 100 g, the frequency is 75 and the class width is 100, so

$$\text{frequency density} = \frac{75}{100} = 0.75$$

Using this scale from the histogram, the frequency densities are:

$$0.75,\ 2.25,\ 1.0,\ 2.5,\ 0.25$$

The total number of books is

$$(100)(0.75) + (100)(2.25) + (250)(1.0) + (50)(2.5) + (200)(0.25)$$

$$= 75 + 225 + 250 + 125 + 50 = 725$$

For weights between 400 g and 600 g:

$$(50)(1.0) + (50)(2.5) + (100)(0.25)$$

$$= 50 + 125 + 25 = 200$$

So the probability is

$$\frac{200}{725} = \frac{8}{29}$$

FINAL ANSWER
$\frac{8}{29}$ or 0.276 to 3 significant figures

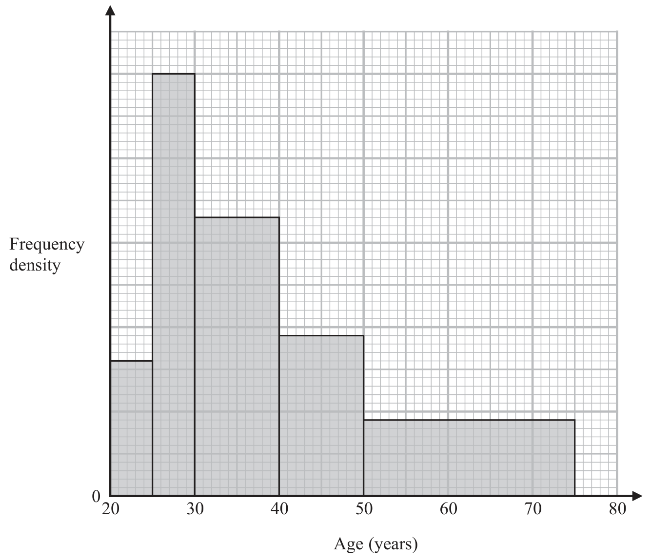
CLASSIFIED PROBLEM

Paper: P1HR | Session: May 2023 | Q#: 22 | Marks: 4

Topic: **Histograms**

Source: May 2023 P1HR

22 Some people attend a concert.
The histogram shows information about the ages of these people.



Work out an estimate for the percentage of these people who are aged more than 55 years.
Give your answer correct to one decimal place.

....%

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 22

Histograms

Topic check. Correct.

Method

The scale of the frequency density axis is not needed because the percentage is found from areas. Using the bar heights from the histogram, the relative areas are:

$$(5)(277) + (5)(866) + (10)(571) + (10)(329) + (25)(155)$$

$$= 1385 + 4330 + 5710 + 3290 + 3875$$

$$= 18590$$

For ages more than 55, use the part of the last bar from 55 to 75:

$$(20)(155) = 3100$$

So the estimated percentage is

$$\frac{3100}{18590} \times 100 = 16.675\ldots$$

FINAL ANSWER

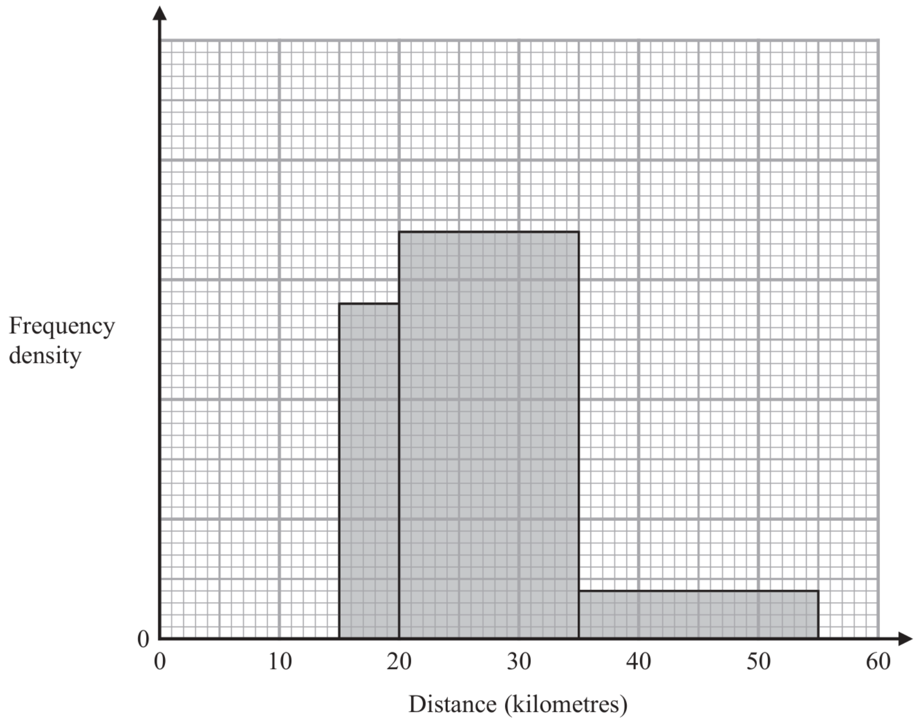
16.7%

CLASSIFIED PROBLEM Paper: P1H Session: May 2024 Q#: 22 Marks: 3

Topic: **Histograms**

Source: May 2024 P1H

22 The incompleted histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.
None of the adults ran more than 55 kilometres.
14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

(Total for Question 22 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 22
Histograms

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $15 \le d < 20$, the frequency is 14 and the class width is 5, so

$$\text{frequency density} = \frac{14}{5} = 2.8$$

Using this scale from the histogram:

$$20 \le d < 35 \quad \text{has frequency density } 3.4$$

$$35 \le d \le 55 \quad \text{has frequency density } 0.4$$

So the known frequencies are

$$14 + (15)(3.4) + (20)(0.4)$$

$$= 14 + 51 + 8 = 73$$

There are 100 adults in total, so the missing frequency from 5 km to 15 km is

$$100 - 73 = 27$$

The missing class width is 10, so its frequency density is

$$\frac{27}{10} = 2.7$$

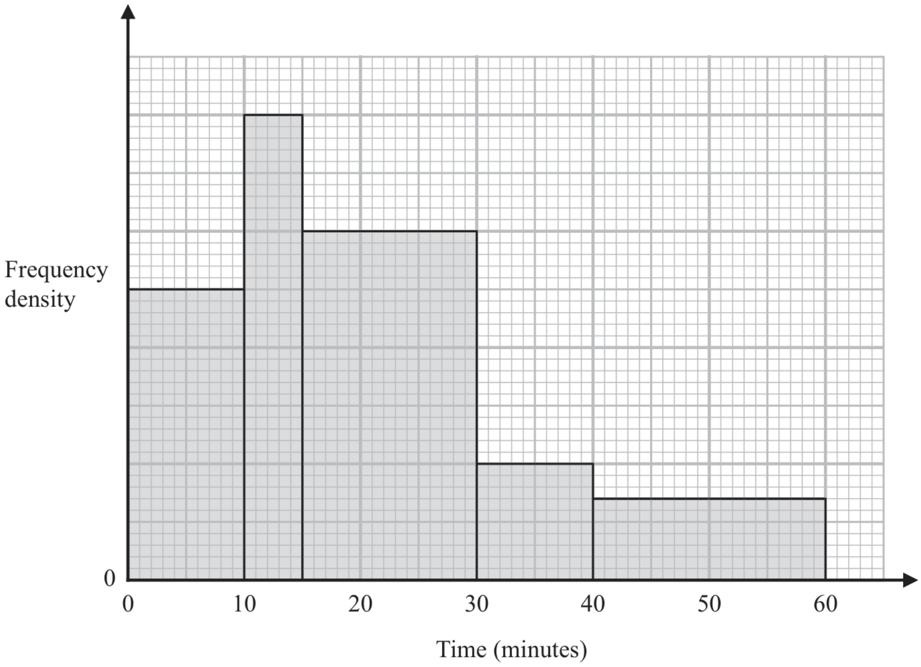
FINAL ANSWER
draw the missing bar from 5 to 15 with frequency density 2.7

CLASSIFIED PROBLEM Paper: P2H | Session: Nov 2025 | Q#: 21 | Marks: 3

TOPIC: **Histograms**

Source: Nov 2025 P2H

21 The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

....

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 21

Histograms

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $10 < t \leq 15$, the frequency is 20 and the class width is 5, so

$$\text{frequency density} = \frac{20}{5} = 4$$

Using this scale from the histogram:

$$15 < t \leq 30 \quad \text{has frequency density } 3$$

$$30 < t \leq 40 \quad \text{has frequency density } 1$$

$$40 < t \leq 60 \quad \text{has frequency density } 0.7$$

For trains at least 25 minutes late:

$$(5)(3) + (10)(1) + (20)(0.7)$$

$$= 15 + 10 + 14$$

$$= 39$$

FINAL ANSWER

39 trains

Topic 23

Combined & Conditional Probability

CLASSIFIED PROBLEM Paper: 4WM1H Session: Nov2025 Q#: 23 Marks: 3

TOPIC: **Combined & Conditional Probability**

Source: Nov 2025 4WM1H

23 There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

....

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 23

Combined & Conditional Probability

Topic check. Correct.

Method

There are 2 pink buttons.
At least one pink button is still in the box unless both pink buttons are taken.
So use the complement.
The probability that both pink buttons are taken is

$$\frac{\binom{2}{2}\binom{13}{1}}{\binom{15}{3}}$$

$$= \frac{13}{455}$$

$$= \frac{1}{35}$$

Therefore,

$$P(\text{at least one pink is still in the box}) = 1 - \frac{1}{35}$$

$$= \frac{34}{35}$$

FINAL ANSWER

$\frac{34}{35}$

Topic 24

Probability Toolkit

CLASSIFIED PROBLEM

Paper: P2HR **Session: Jan 2021** **Q#: 21** **Marks: 6**

TOPIC: **Probability Toolkit**

Source: Jan 2021 P2HR

21 A bag contains n beads.
6 of the beads are red and the rest are blue.

Ravi is going to take at random 2 beads from the bag.

The probability that the 2 beads will be of the same colour is $\frac{9}{17}$

Using algebra, and showing each stage of your working, calculate the value of n.

$n =$ ....

(Total for Question 21 is 6 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jan 2021 P2HR | Question 21
Probability Toolkit

Topic check. Corrected from Right-Angled Triangles to Probability Toolkit.

Method
There are n beads altogether.
Red beads:

$$6$$

Blue beads:

$$n-6$$

The probability of taking two beads of the same colour is

$$P(RR)+P(BB)=\frac{9}{17}$$

So

$$\frac{6}{n}\cdot\frac{5}{n-1}+\frac{n-6}{n}\cdot\frac{n-7}{n-1}=\frac{9}{17}$$

$$\frac{30+(n-6)(n-7)}{n(n-1)}=\frac{9}{17}$$

Expand:

$$(n-6)(n-7)=n^2-13n+42$$

So

$$\frac{n^2-13n+72}{n(n-1)}=\frac{9}{17}$$

Cross multiply:

$$17(n^2-13n+72)=9n(n-1)$$

$$17n^2-221n+1224=9n^2-9n$$

$$8n^2-212n+1224=0$$

Divide by 4:

$$2n^2-53n+306=0$$

Factor:

$$(2n-17)(n-18)=0$$

So

$$n = \frac{17}{2} \quad \text{or} \quad n = 18$$

The number of beads must be a whole number, so

$$n = 18$$

FINAL ANSWER

$\boxed{18}$

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jun 2022 | Q#: 24 | Marks: 5

TOPIC: **Probability Toolkit**

Source: Jun 2022 P1HR

24 Elliot has x counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.
He then picks at random a second counter and turns the counter over.

The probability that there are still 5 counters showing a red face is $\frac{19}{32}$

Work out the value of x
Show clear algebraic working.

$x =$ ....

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 24

Probability Toolkit

Topic check. Correct.

Method

There are x counters. Initially 5 show red, so $x - 5$ show green.
After two turns, there are still 5 red faces if:

- the same counter is picked twice, or
- one red counter and one green counter are picked.

The probability of picking the same counter twice is

$$\frac{1}{x}$$

The probability of picking one red and one green in either order is

$$\frac{5}{x} \cdot \frac{x-5}{x} + \frac{x-5}{x} \cdot \frac{5}{x}$$

$$= \frac{10(x-5)}{x^2}$$

So

$$\frac{1}{x} + \frac{10(x-5)}{x^2} = \frac{19}{32}$$

$$\frac{11x-50}{x^2} = \frac{19}{32}$$

$$32(11x-50) = 19x^2$$

$$19x^2 - 352x + 1600 = 0$$

$$(x-8)(19x-200) = 0$$

Since x must be a whole number,

$$x = 8$$

FINAL ANSWER

$x = 8$

CLASSIFIED PROBLEM

Paper: P1H | Session: Jun 2025 | Q#: 21 | Marks: 3

TOPIC: **Probability Toolkit**

Source: Jun 2025 P1H

21 A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

....

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 21
Probability Toolkit

Topic check. Correct.

Method

There are

$$\binom{20}{3} = 1140$$

ways to choose 3 counters.
Exactly two counters are the same colour:

$$\binom{9}{2}(11) + \binom{7}{2}(13) + \binom{4}{2}(16)$$

$$= 36(11) + 21(13) + 6(16)$$

$$= 396 + 273 + 96 = 765$$

So the probability is

$$\frac{765}{1140} = \frac{51}{76}$$

FINAL ANSWER
$\frac{51}{76}$

CLASSIFIED PROBLEM

Paper: P2H | Session: May 2021 | Q#: 23 | Marks: 5

TOPIC: **Probability Toolkit**

Source: May 2021 P2H

23 Pippa has a box containing N pens.

There are only black pens and red pens in the box.
The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$

Find the possible values of N.
Show clear algebraic working.

....

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P2H | **Question 23**
Probability Toolkit

Topic check. Correct.

Method

Let the number of red pens be r.
Then the number of black pens is

$$r + 3$$

and

$$N = 2r + 3$$

So

$$r = \frac{N-3}{2}$$

and

$$r + 3 = \frac{N+3}{2}$$

The probability of black followed by red is

$$\frac{(N+3)/2}{N} \cdot \frac{(N-3)/2}{N-1} = \frac{9}{35}$$

$$\frac{N^2-9}{4N(N-1)} = \frac{9}{35}$$

$$35(N^2-9) = 36N(N-1)$$

$$35N^2 - 315 = 36N^2 - 36N$$

$$N^2 - 36N + 315 = 0$$

$$(N-15)(N-21) = 0$$

FINAL ANSWER
$N = 15$ or $N = 21$

CLASSIFIED PROBLEM Paper: P1HR | Session: May 2024 | Q#: 21 | Marks: 5

TOPIC: **Probability Toolkit**

Source: May 2024 P1HR

21 There are 25 counters in a bag such that

6 counters are blue
x counters are orange, where $x > 9$
the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is $\frac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.
Show clear algebraic working.

....

(Total for Question 21 is 5 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

May 2024 P1HR | Question 21
Probability Toolkit

Topic check. Correct.

Method
There are 25 counters.
Blue counters:

$$6$$

Orange counters:

$$x$$

Pink counters:

$$25 - 6 - x = 19 - x$$

The probability of taking one orange and one pink in either order is

$$\frac{x}{25} \cdot \frac{19-x}{24} + \frac{19-x}{25} \cdot \frac{x}{24}$$

$$= \frac{2x(19-x)}{25 \cdot 24}$$

This is equal to $\frac{22}{75}$, so

$$\frac{2x(19-x)}{600} = \frac{22}{75}$$

$$\frac{x(19-x)}{300} = \frac{22}{75}$$

$$x(19-x) = 88$$

$$19x - x^2 = 88$$

$$x^2 - 19x + 88 = 0$$

$$(x-8)(x-11) = 0$$

Since $x > 9$,

$$x = 11$$

So the number of pink counters is

$$19 - 11 = 8$$

The probability of taking 2 pink counters is

$$\frac{8}{25} \cdot \frac{7}{24} = \frac{56}{600}$$

$$= \frac{7}{75}$$

FINAL ANSWER

$\frac{7}{75}$

CLASSIFIED PROBLEM | Paper: P1HR | Session: MayNov 2020 | Q#: 23 | Marks: 4

TOPIC: **Probability Toolkit**

Source: May-Nov 2020 P1HR

23 In a bag, there are only

3 blue beads
4 white beads
and x orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is $\frac{3}{8}$

Find the total number of beads in the bag.
Show clear algebraic working.

....

(Total for Question 23 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | Question 23
Probability Toolkit

Topic check. Correct.

Method
The total number of beads is

$$x + 7$$

The probability of choosing two beads of the same colour is

$$\frac{\binom{3}{2} + \binom{4}{2} + \binom{x}{2}}{\binom{x+7}{2}}$$

This is equal to $\frac{3}{8}$, so

$$\frac{3 + 6 + \frac{x(x-1)}{2}}{\frac{(x+7)(x+6)}{2}} = \frac{3}{8}$$

$$\frac{x^2 - x + 18}{(x+7)(x+6)} = \frac{3}{8}$$

$$8(x^2 - x + 18) = 3(x+7)(x+6)$$

$$8x^2 - 8x + 144 = 3x^2 + 39x + 126$$

$$5x^2 - 47x + 18 = 0$$

$$(x-9)(5x-2) = 0$$

Since x must be a whole number,

$$x = 9$$

So the total number of beads is

$$9 + 7 = 16$$

FINAL ANSWER
16 beads

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2023 | Q#: 20 | Marks: 4

TOPIC: **Probability Toolkit**

Source: Nov 2023 P1H

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag. He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

....

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P1H | Question 20

Probability Toolkit

Topic check. Moved to Probability Toolkit.

Method

Let the probability of taking a 10 cent coin be p.
The mean is 10 cents only when both coins are 10 cent coins.

$$p^2 = \frac{49}{121}$$

$$p = \frac{7}{11}$$

So the probability of taking a 20 cent coin is

$$1 - \frac{7}{11} = \frac{4}{11}$$

The mean is 15 cents when one coin is 10 cents and the other is 20 cents.
There are two possible orders:

$$P(10, 20) + P(20, 10)$$

$$= \frac{7}{11} \cdot \frac{4}{11} + \frac{4}{11} \cdot \frac{7}{11}$$

$$= \frac{56}{121}$$

FINAL ANSWER

$\frac{56}{121}$

Topic 25

Probability Diagrams - Venn & Tree Diagrams

CLASSIFIED PROBLEM Paper: P2H Session: Jun 2025 Q#: 20 Marks: 6

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Source: Jun 2025 P2H

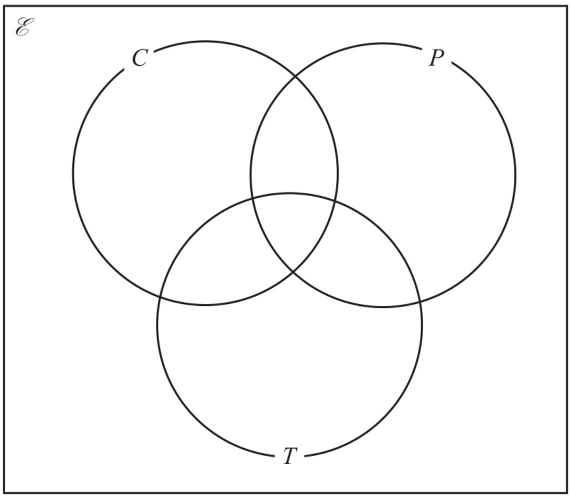
20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots
12 grow carrots and potatoes and tomatoes
18 grow carrots and potatoes
27 grow carrots and tomatoes
32 grow potatoes and tomatoes
29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

....

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

....

(2)

(Total for Question 20 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 20

Probability Diagrams - Venn & Tree Diagrams

Topic check. Correct.

Method

There are 120 gardeners in total and 29 are outside all three sets, so

$$n(C \cup P \cup T) = 120 - 29 = 91$$

The triple intersection is

$$n(C \cap P \cap T) = 12$$

So the pair-only regions are:

$$C \cap P \text{ only} = 18 - 12 = 6$$

$$C \cap T \text{ only} = 27 - 12 = 15$$

$$P \cap T \text{ only} = 32 - 12 = 20$$

Since 43 grow carrots,

$$C \text{ only} = 43 - 6 - 15 - 12 = 10$$

Let the number who grow only potatoes be y. The number who grow only tomatoes is also y. Now add the regions in the union:

$$10 + y + y + 6 + 15 + 20 + 12 = 91$$

$$63 + 2y = 91$$

$$2y = 28$$

$$y = 14$$

So the Venn regions are:

$$C \text{ only} = 10, \quad P \text{ only} = 14, \quad T \text{ only} = 14$$

$$C \cap P \text{ only} = 6, \quad C \cap T \text{ only} = 15, \quad P \cap T \text{ only} = 20$$

$$C \cap P \cap T = 12, \quad \text{outside} = 29$$

For part (b),

$$n(T' \cap C) = C \text{ only} + C \cap P \text{ only}$$

$$= 10 + 6 = 16$$

For part (c), one gardener who grows carrots is chosen.

$$P(\text{also grows potatoes}) = \frac{n(C \cap P)}{n(C)}$$

$$= \frac{18}{43}$$

FINAL ANSWER

(b) 16, (c) $\frac{18}{43}$

Topic 26

Combined & Conditional Probability

CLASSIFIED PROBLEM Paper: P2H | Session: Jun 2023 | Q#: 20 | Marks: 3

TOPIC: **Combined & Conditional Probability**

Source: Jun 2023 P2H

20 There are 12 counters in a bag.

3 of the counters are red
9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

....

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 20

Combined & Conditional Probability

Topic check. Correct.

Method

There are 3 red counters.
At least one red counter is still in the bag unless all 3 red counters are taken.
So use the complement:

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

$$= \frac{6}{1320}$$

$$= \frac{1}{220}$$

Therefore,

$$P(\text{at least one red is still in the bag}) = 1 - \frac{1}{220}$$

$$= \frac{219}{220}$$

FINAL ANSWER

$\frac{219}{220}$

CLASSIFIED PROBLEM Paper: P1H Session: Nov 2025 Q#: 22 Marks: 3

TOPIC: **Combined & Conditional Probability**

Source: Nov 2025 P1H

22 There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

....

(Total for Question 22 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 22

Combined & Conditional Probability

Topic check. Correct.

Method

There are 2 pink buttons.
At least one pink button is still in the box unless both pink buttons are taken.
So use the complement.
The probability that both pink buttons are taken is

$$\frac{\binom{2}{2}\binom{13}{1}}{\binom{15}{3}}$$

$$= \frac{13}{455}$$

$$= \frac{1}{35}$$

Therefore,

$$P(\text{at least one pink is still in the box}) = 1 - \frac{1}{35}$$

$$= \frac{34}{35}$$

FINAL ANSWER

$\frac{34}{35}$